Neuroscience

Humans underestimate their body mass in microgravity: evidence from reaching movements during spaceflight

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eLife Assessment

This paper undertakes an **important** investigation to determine whether movement slowing in microgravity is due to a strategic conservative approach or rather due to an underestimation of the mass of the arm. While the experimental dataset is unique and the coupled experimental and computational analyses comprehensive, the authors present **incomplete** results to support the claim that movement slowing is due to mass underestimation. Further analysis is needed to rule out alternative explanations.

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Abstract

Astronauts consistently exhibit slower movements in microgravity, even during tasks requiring rapid responses. The sensorimotor mechanisms underlying this general slowing remain debated. Two competing hypotheses have been proposed: either the sensorimotor system adopts a conservative control strategy for safety and postural stability, or the system underestimates body mass due to reduced inputs from proprioceptive receptors. To resolve the debate, we studied twelve taikonauts aboard the China Space Station performing a classical hand-reaching task. Compared to their pre-flight performance and to an age-matched control group, participants showed increased movement durations and altered kinematic profiles in microgravity. Model-based analyses of motor control parameters revealed that these changes stemmed from reduced initial force generation in the feedforward control phase followed by compensatory feedback-based corrections. These findings support the body mass underestimation hypothesis while refuting the strategic slowing hypothesis. Importantly, sensory estimate of bodily property in microgravity is biased but evaded from sensorimotor adaptation, calling for an extension of existing theories of motor learning.

Introduction

Successful space exploration fundamentally depends on efficient sensorimotor control under microgravity conditions, where both movement accuracy and speed are critical (Strangman et al., 2014). A consistently observed phenomenon is that astronauts exhibit slower movements during routine activities in space stations (Kubis et al., 1977). This general movement slowing persists even in controlled experimental settings where astronauts are securely fastened without concerns for postural stability or safety. Indeed, fast movements in microgravity can be disadvantageous, potentially destabilizing posture, complicating the arrest of accelerated body segments, and increasing collision risks in confined spacecraft environments (Bock, 1998; Layne et al., 2001; Paloski et al., 1999; Reschke et al., 1998; Tafforin et al., 1989). Yet even when body stability is ensured, well-practiced goal-directed actions consistently show prolonged movement durations across various tasks, from hand pointing to visual targets (Berger et al., 1997; Bock et al., 2001, 2003; Mechtcheriakov et al., 2002) to step tracking with joysticks (Newman & Lathan, 1999; Sangals et al., 1999). This slowing manifests as reduced peak speeds in hand pointing (Mechtcheriakov et al., 2002; Sangals et al., 1999; Weber & Stelzer, 2022). Despite extensive documentation of this phenomenon, the underlying sensorimotor mechanisms remain unclear (Bock et al., 1996b; Weber et al., 2019), presenting a significant challenge for optimizing human performance in space.

Two primary hypotheses have emerged to explain the slowing of goal-directed actions in microgravity. The conservative control hypothesis suggests that the sensorimotor system adopts a generalized strategy prioritizing safety and postural stability over speed (Berger et al., 1997; Bock et al., 2010; Mechtcheriakov et al., 2002). This adaptation may arise from the novel challenges of movement control in microgravity, where even simple actions can propagate unexpected forces through body segments. The persistent application of this conservative strategy, as a generalized adaptation to the novel environment, could explain the movement slowing in microgravity. Alternatively, the body mass underestimation hypothesis builds on observations that humans consistently underestimate the mass of handheld objects in microgravity (Ross & Reschke, 1982), suggesting that similar perceptual biases might extend to the estimation of body segment properties (Bock et al., 1996b). Such misperception would directly impact the feedforward control of movement, potentially leading to systematic under-actuation of initial movements (Elliott et al., 2010, 2017).

Both hypotheses predict reduced peak speed and acceleration in microgravity, but they diverge in their predictions about movement kinematics. First, typical reaching movement exhibits a symmetrical bell-shaped speed profile, which minimizes energy expenditure while maximizing accuracy according to optimal control principles (Todorov, 2004). The conservative control hypothesis suggests a maintained temporal symmetry as longer durations are strategically planned for optimal performance, the peak speed and acceleration are thus delayed in time (**Figure 1A**). In contrast, the body mass underestimation hypothesis predicts that insufficient initial force generation, resulting from a systematic sensory bias in internal models that predict movement dynamics, would lead to earlier occurrence of peak speed and acceleration (Elliott et al., 2010, 2017). Previously, the timing of these peaks has shown mixed results, with some studies reporting earlier peaks (Sangals et al., 1999), others finding delays (Fowler et al., 2008), and some showing no temporal shifts (Berger et al., 1997; Mechtcheriakov et al., 2002). Second, the hypotheses make distinct predictions about feedback-based corrections during late reaching. The conservative control hypothesis suggests that movements are optimally adapted for the microgravity environment, thus requiring no additional corrections. Instead, the mass underestimation hypothesis predicts that initial underactuation would necessitate subsequent feedback-based corrections to reach the target (Chua & Elliott, 1993; Smeets & Brenner, 1995). These corrections would manifest as additional submovements, contributing to asymmetric speed profiles (**Figure 1A**). Current evidence for increased feedback-based corrections in microgravity remains inconsistent: while some studies support their presence (Sangals et al., 1999), others show conflicting results (Mechtcheriakov et al., 2002) or opposite effects (Fowler et al., 2008). If present, these corrective submovements should predict movement duration prolongation, serving as crucial evidence for the mass-underestimation account for the general slowing in microgravity.

To resolve the theoretical debate, we leveraged a unique biomechanical property of arm movements: the natural variation in effective limb mass across movement directions, known as anisotropic inertia (Mussa-Ivaldi et al., 1985). Based on a two-joint arm model (Li & Todorov, 2004), reaching movements toward targets in the frontal-medial direction involve greater effective mass compared to those in the frontal-lateral direction. Based on the motor utility theory, both the magnitude and timing of peak speed and acceleration should be systematically modulated by target direction (Shadmehr et al., 2016). Similarly, if astronauts underestimate their limb mass by approximately 30% in microgravity (Crevecoeur et al., 2014), we would expect reduced peak speed/acceleration and earlier occurrence of these peaks across all the directions. Here we examined twelve taikonauts before, during, and after their 3- or 6-month missions on the China Space Station (CSS) and compared their performance to age-matched ground controls. Participants performed rapid reaching movements to targets specifically chosen to engage different effective masses. Our approach allows us to dissociate the effects of mass underestimation from strategic control by examining how movement kinematics vary with direction-dependent effective mass.

Results

We compared the performance of 12 taikonauts to that of age-matched ground controls across multiple test sessions (**Figure 6B** and Table S1). Participants performed rapid reaching movements toward one of three targets, presented in pseudorandom order to minimize movement automation and ensure active planning for each reach. The three target directions (45°, 90°, and 135°) were chosen to vary the effective mass of the moving limb, as verified through biomechanical simulations based on the two-joint arm model (Li & Todorov, 2004; **Figure 1B**). Our model simulations not only demonstrated the direction-dependent variation in limb mass but also predicted how actual and misperceived limb mass would affect reaching kinematics (**Figure 1C-H**). These predictions were derived by combining the movement utility framework for estimating planned movement duration (Shadmehr et al., 2016; **Figure S1A-B**) with a forward optimal controller for simulating reaching trajectories (Todorov, 2005; **Figure S1C-D**). Notably, the model predicted that underestimating limb mass would lead to reduced and earlier peak speed/acceleration across directions (**Figure 1C-H**).

Movement Accuracy and Reaction Time Remain Intact in Microgravity

All participants successfully completed the task. Invalid trials, comprising early initiation (RT < 150 ms), late initiation (RT > 400 ms), or measurement failures, accounted for only 1.87% of trials in the experimental group and 0.70% in the control group. The percentage of invalid trials remained stable across the pre-in and post-flight phases (Friedman's tests, all $p > 0.1$).

Reaching accuracy remained high despite the stringent speed requirement of completing movements within 650 ms after target appearance (**Figure S2**). Analysis of endpoint error using a 3 (target direction) × 3 (phase) repeated-measures ANOVA revealed no significant differences across phases for either group. While the experimental group showed slightly better accuracy for the 45° target, this difference was minimal (0.04 cm over a 12 cm reach distance).

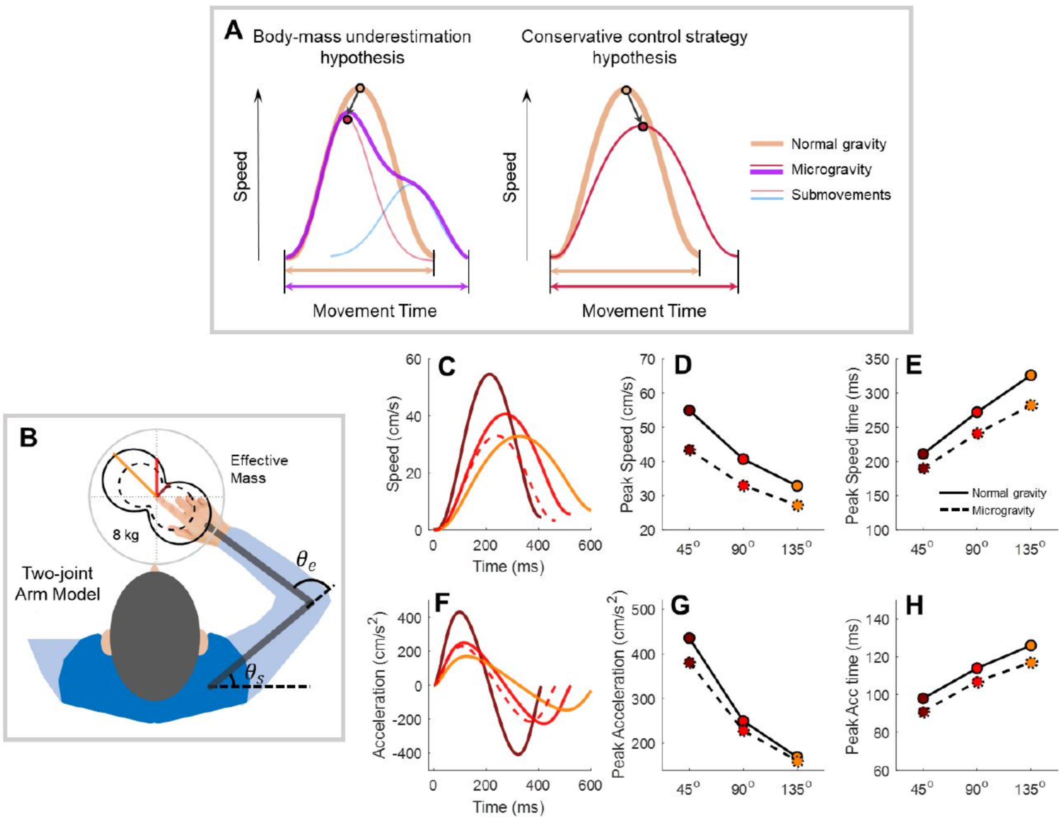


Figure 1.

Model simulations revealing characteristic kinematic features of reaching movements.

A) Speed profile predictions under two competing hypotheses. Under normal gravity, hand speed follows a typical profile (bold apricot lines). According to the body mass underestimation hypothesis (left panel), initial underactuation produces a lower, earlier peak speed (light red), necessitating a corrective submovement (light blue) that results in an asymmetric, prolonged profile (purple). The conservative strategy hypothesis (right panel) predicts symmetric slowing with delayed peak speed (red). **B)** Two-joint arm model and its simulated effective mass. The spatial distribution of effective mass is shown by black curves (solid: normal gravity; dashed: 30% mass underestimation in microgravity). Colored lines indicate effective mass values for our three experimental target directions (45°, 90°, 135°). **C & F)** Representative speed and acceleration profiles simulated using effective masses from B. Profiles shown for normal (solid) and underestimated (dashed) body mass in the 90° direction demonstrate reduced, earlier peaks with mass underestimation. **D-E & G-H)** Simulated kinematic parameters across movement directions under normal (solid) and underestimated (dashed) mass conditions. Mass underestimation consistently leads to reduced amplitude and earlier timing of speed/acceleration peaks across all directions. Model simulation details provided in Supplementary Note 1.

In half of the trials, a beep accompanied target appearance to promote faster reactions, allowing us to examine speed-accuracy trade-offs during action planning (Sutter et al., 2021). The beep effectively reduced median reaction times by approximately 38 ms (242 ± 49 ms vs. 281 ± 50 ms; **Figure 2A**; see control group data in **Figure S3A**). To evaluate microgravity effects, we pooled data from both beep conditions and conducted a 3 (direction) × 3 (phase) repeated-measures ANOVA. The experimental group showed significant main effects of direction ($F(2,22) = 25.260$, $p < 0.001$, partial $\eta^2 = 0.697$) and phase ($F(2,22) = 5.820$, $p = 0.014$, partial $\eta^2 = 0.346$), without interaction ($F(4,44) = 2.246$, $p = 0.111$). Post-hoc analyses revealed slower reaction times for the 45° direction compared to both 90° ($p < 0.001$, d = 0.293) and 135° ($p = 0.003$, d = 0.284). Notably, reactions were faster during the in-flight phase compared to pre-flight ($p = 0.037$, d = 0.333), with no significant difference between in-flight and post-flight phases ($p = 0.127$). Similar effects were observed in the control group (**Figure S3A**), suggesting that the reduced reaction times likely reflect practice effects rather than microgravity influence. Thus, participants maintained, and even improved, their ability to rapidly initiate movements in microgravity with reaching accuracy.

Microgravity leads to slower movement

Although showing no decrement in movement accuracy or RT, the taikonauts exhibited prolonged movement durations (MD) during the in-flight phase compared to both pre- and post-flight phases (**Figure 2B**). A two-way repeated-measures ANOVA reveals significant main effects of direction ($F(2,22) = 62.555$, $p < 0.001$, partial $\eta^2 = 0.850$) and phase ($F(2,22) = 6.761$, $p = 0.015$, partial $\eta^2 = 0.381$), with no significant interaction ($F(4,44) = 1.413$, $p = 0.263$). Post-hoc comparisons showed that MD was significantly prolonged during the in-flight phase (vs. pre-flight: $p = 0.014$, d = 0.333; vs. post-flight: $p = 0.013$, d = 0.181). MD was also significantly greater for the 90° and 135° directions—those associated with larger effective mass—compared to the 45° direction (all $p < 0.001$ in pairwise comparisons). In contrast, the control group showed a similar directional effect but, importantly, did not show a phase effect (**Figure S3B**), suggesting that movement slowing is specific to spaceflight.

Microgravity leads to reduced peak acceleration/speed and advanced timing of these peaks

The initial movement exhibited signs of underactuation during spaceflight, consistent with the body mass underestimation hypothesis. Peak acceleration and peak speed, established measures of feedforward control (Elliott et al., 2010, 2017), both showed significant decreases during the in-flight phase (**Figure 3**). For peak acceleration, there were significant main effects of phase ($F(2,22) = 6.401$, $p = 0.015$, partial $\eta^2 = 0.368$) and direction ($F(2,22) = 75.516$, $p < 0.001$, partial $\eta^2 = 0.873$), along with an interaction effect ($F(4,44) = 4.548$, $p = 0.016$, partial $\eta^2 = 0.293$). Consistent with model simulations, the 45° direction, associated with a lower effective mass, showed significantly higher peak acceleration than the other two directions (**Figure 3A**; all $p < 0.001$ in pairwise comparisons). Peak acceleration decreased significantly during spaceflight across all three directions (simple main effects: 45°, $p = 0.020$, $\eta^2 = 0.543$; 90°, $p = 0.004$, $\eta^2 = 0.673$; 135°, $p = 0.005$, $\eta^2 = 0.657$). Planned contrasts further indicated lower peak acceleration in the in-flight phase compared to pre-flight (45°, $p = 0.006$, $\eta^2 = 0.516$; 90°, $p = 0.008$, $\eta^2 = 0.485$; 135°, $p = 0.003$, $\eta^2 = 0.560$), with marginal recovery in the post-flight phase for two directions (45°, $p = 0.071$, $\eta^2 = 0.267$; 90°, $p = 0.062$, $\eta^2 = 0.282$) and significant recovery for one direction (135°, $p = 0.029$, $\eta^2 = 0.363$).. In contrast, the control group displayed only the direction effect (**Figure S3C-D**), confirming that the underactuation was specific to microgravity.

Peak speed showed parallel changes during spaceflight (**Figure 3B**). Analysis revealed significant main effects of phase ($F(2,22) = 7.043$, $p = 0.009$, partial $\eta^2 = 0.390$) and direction ($F(2,22) = 114.420$, $p < 0.001$, partial $\eta^2 = 0.912$), without interaction ($F(4,44) = 1.929$, $p = 0.161$). Peak speed was significantly reduced during the in-flight phase compared to both pre-flight ($p = 0.004$, d = 0.331) and post-flight phases ($p = 0.046$, d = 0.199), with evidence of recovery post-flight (pre-vs.

Figure 2.

Reaction time and movement duration in the experimental group.

A) Median reaction times plotted by experimental phase (x-axis) and target direction (different colors). Dashed lines and dot lines represent the "beep" and "no beep" conditions. **B)** Average movement durations shown in the same format. Data are combined across beep conditions due to no significant differences between them. In both panels, error bars denote standard error across participants. Asterisks indicate significance levels for post-hoc phase comparisons (*p < 0.05, **p < 0.01, ***p < 0.001).

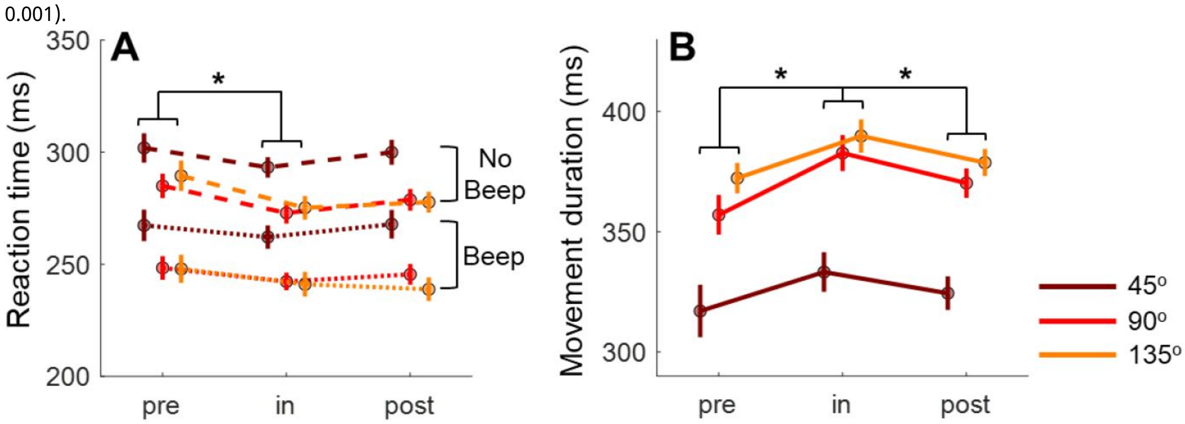
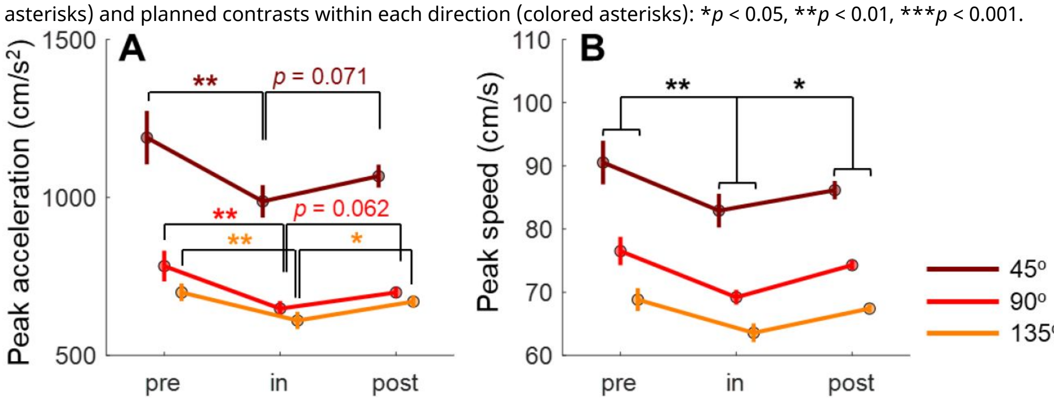


Figure 3.

Magnitude changes of peak acceleration and speed during spaceflight.

A) Peak acceleration across experimental phases and movement directions, showing systematic decreases during the in-flight phase. **B)** Peak speed data presented in the same format, demonstrating parallel changes to peak acceleration. Error bars indicate standard error across participants. Statistical significance levels are shown for both overall phase comparisons (black asterisks) and planned contrasts within each direction (colored asterisks): *p < 0.05, **p < 0.01, ***p < 0.001.



post-flight: $p = 0.483$, d = 0.132). Target direction systematically modulated peak speed (all pairwise comparisons, $p < 0.001$), with the 45° direction showing highest speeds and 135° lowest—matching predictions based on the movement utility theory. The control group showed only direction effects without phase-related changes (**Figure S3D**), confirming the specificity of these changes to microgravity exposure.

The critical test for the competing hypotheses of movement slowing lies in whether peak acceleration and peak speed occur earlier or later in microgravity. The body mass underestimation hypothesis predicts earlier timing, whereas the conservative control strategy hypothesis predicts delayed timing. We observed an overall timing advance of these two measures. For peak acceleration time in the experimental group (**Figure 4A**), two-way repeated-measures ANOVAs revealed significant main effects of phase ($F(2,22) = 5.189$, $p = 0.024$, partial $\eta^2 = 0.321$) and direction ($F(2,22) = 30.375$, $p < 0.001$, partial $\eta^2 = 0.734$), along with an interaction effect ($F(4,44) = 4.503$, $p = 0.013$, partial $\eta^2 = 0.290$). In line with predictions, peak acceleration appeared significantly earlier in the 45° direction than other directions (45° vs. 90°, $p < 0.001$, d = 0.304; 45° vs. 135°, $p < 0.001$, d = 0.271). The phase effect emerged only for directions with higher effective mass (simple main effects, 90°: $F(2,10) = 6.660$, $p = 0.015$, $\eta^2 = 0.571$; 135°: $F(2,10) = 8.411$, $p = 0.007$, $\eta^2 = 0.627$). Planned contrasts confirmed earlier peak acceleration during spaceflight compared to post-flight (90°, $p = 0.005$, $\eta^2 = 0.530$; 135°, $p = 0.004$, $\eta^2 = 0.537$), with marginal differences versus pre-flight (135°, $p = 0.065$). The control group showed only directional effects without phase effect (**Figure S3E**).

Given that overall movement duration is prolonged in microgravity, a more direct way to assess the symmetry of the speed profile is to compute the relative timing of speed/acceleration peaks as a percentage of MD. Indeed, the relative timing of peak acceleration showed the same reduction in microgravity with a more pronounced effect than absolute timing (**Figure 4C**). A two-way repeated-measures ANOVA revealed significant main effects of phase ($F(2,22) = 15.414$, $p < 0.001$, partial $\eta^2 = 0.584$) and interaction ($F(2,22) = 7.223$, $p = 0.002$, partial $\eta^2 = 0.396$), while the main effect of direction was marginally significant ($F(2,22) = 3.374$, $p = 0.053$, partial $\eta^2 = 0.235$). Phase effects were significant for directions with higher effective mass (simple main effects: 90°, $F(2,10) = 13.327$, $p = 0.002$, $\eta^2 = 0.727$; 135°, $F(2,10) = 24.574$, p 0.001, $\eta^2 = 0.831$) but not for 45° ($F(2,10) = 0.609$, $p = 0.563$). Planned contrasts confirmed significantly earlier timing during spaceflight compared to both pre- and post-flight phases (all $p < 0.002$ for 90° and 135° directions). None of these effects appeared in the control group (**Figure S3G**).

Peak speed also showed a timing advance, but only for relative timing. For absolute timing (**Figure 4B**), a two-way repeated-measures ANOVA revealed a significant main effect of direction ($F(2,22) = 57.428$, $p < 0.001$, partial $\eta^2 = 0.839$) and a marginal interaction ($F(4,44) = 2.233$, $p = 0.095$, partial $\eta^2 = 0.169$), without a phase effect ($F(2,22) = 0.093$, $p = 0.864$). In contrast, relative timing analysis (**Figure 4D**) showed significant effects of phase ($F(2,22) = 8.941$, $p = 0.004$, partial $\eta^2 = 0.448$), direction ($F(2,22) = 4.991$, $p = 0.030$, partial $\eta^2 = 0.312$), and their interaction ($F(4,44) = 4.755$, $p = 0.015$, partial $\eta^2 = 0.302$). The phase effect appeared only for directions with higher effective mass (90°: $F(2,10) = 17.278$, $p = 0.001$, $\eta^2 = 0.776$; 135°: $F(2,10) = 6.189$, $p = 0.018$, $\eta^2 = 0.553$). Planned contrasts revealed significantly earlier relative timing during spaceflight versus pre-flight for both 90° ($p < 0.001$, $\eta^2 = 0.770$) and 135° ($p = 0.007$, $\eta^2 = 0.496$) directions, with significant post-flight recovery in 90° ($p = 0.003$, $\eta^2 = 0.575$). The control group showed only directional effects (**Figure S3F** & **2H**). Thus, after accounting for the prolongation of movement duration, we observed a temporal advance of peak acceleration and speed during spaceflight.

Movement slowing is associated with corrective submovements

The feedforward component of movement, captured by our model simulations, accounts for only the initial phase of reaching. With peak acceleration and speed occurring at approximately 80 ms and 150 ms after movement onset, and total movement duration spanning 300-400 ms, participants had ample opportunity for feedback-based corrections (Dimitriou et al., 2013).

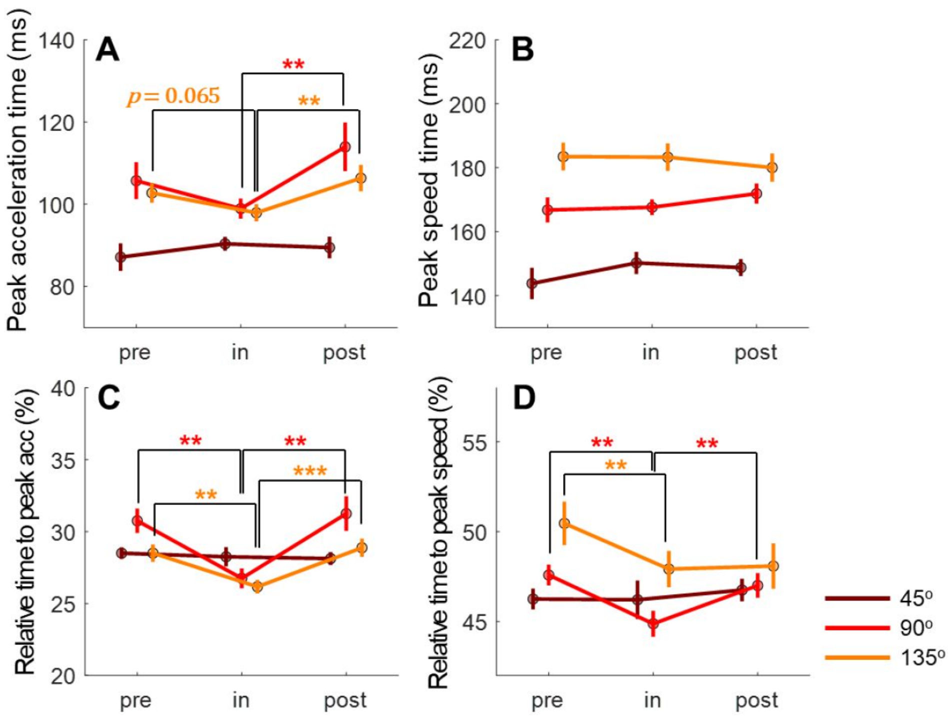


Figure 4.

Temporal analysis of peak acceleration and peak speed during spaceflight.

A-B) Time to peak acceleration and peak speed across experimental phases and movement directions. **C-D)** Relative timing analysis showing these same peaks as a percentage of total movement duration, revealing more pronounced microgravity effects than absolute timing measures. Error bars indicate standard error across participants. Asterisks denote significance levels for planned phase comparisons (**p* < 0.05, ***p* < 0.01, ****p* < 0.001). Note the systematic modulation of timing by phase, particularly for high effective mass conditions.

These corrections, if present, would manifest as secondary submovements (Meyer et al., 1988). While optimal reaching movements typically show symmetrical speed profiles (Flash & Hogan, 1985), the presence of corrective submovements creates characteristic left-skewed speed profiles (Elliott et al., 2010; Meyer et al., 1988; Woodworth, 1899). To identify such corrections, we employed an established decomposition method (Rohrer & Hogan, 2006) to detect trials containing both primary and secondary submovements (two-peak trials; **Figure 5A**).

Analysis revealed a striking increase in corrective movements during spaceflight. The percentage of trials showing two distinct submovements increased significantly during the in-flight phase (**Figure 5B**), evidenced by a main effect of phase ($F(2,22) = 13.969$, $p < 0.001$, partial $\eta^2 = 0.559$) and a marginal phase-by-direction interaction ($F(2,22) = 2.642$, $p = 0.054$, partial $\eta^2 = 0.194$). This effect was specific to microgravity exposure, as the control group showed no significant changes across phases (all $p > 0.05$; **Figure S4A**). Notably, the increase in corrective movements was most pronounced for directions with higher effective mass (simple main effects: 90°, $F(2,10) = 11.426$, $p = 0.003$, $\eta^2 = 0.696$; 135°, $F(2,10) = 7.292$, $p = 0.011$, $\eta^2 = 0.593$).

To understand how feedback corrections contribute to movement slowing, we analyzed the inter-peak interval (IPI) between primary and corrective submovements, an established measure of feedback-based control (Craik, 1947; Meyer et al., 1988, 2018). The taikonauts showed markedly longer IPIs during spaceflight (**Figure 5C**), with a two-way ANOVA revealing significant effects of phase ($F(2,22) = 11.150$, $p = 0.001$, partial $\eta^2 = 0.503$), direction ($F(2,22) = 4.721$, $p = 0.026$, partial $\eta^2 = 0.300$), and their interaction ($F(2,22) = 6.559$, $p = 0.002$, partial $\eta^2 = 0.374$). Critically, IPI increases were confined to directions with greater effective mass (90°, $F(2,10) = 11.573$, $p = 0.002$, $\eta^2 = 0.698$; 135°, $F(2,10) = 13.371$, $p = 0.001$, $\eta^2 = 0.728$). The control group exhibited only directional effects ($F(2,22) = 19.277$, $p < 0.001$, partial $\eta^2 = 0.637$; **Figure S4B**) without phase-related changes (all $p > 0.13$).

To establish whether these enlarged corrective movements explain the overall movement slowing, we examined how changes in IPI (ΔIPI) predict changes in movement duration (ΔMD) during spaceflight (**Figure 5D**). A Linear Mixed Model analysis incorporating ΔIPI, phase transition (pre-to-in and post-to-in), and direction revealed that ΔIPI significantly predicted ΔMD ($\beta = 0.458$ [0.144, 0.771], $t(67) = 2.915$, $p = 0.005$, partial-$R^2 = 0.103$). Phase transition also showed significance ($\beta = -0.01$ [-0.017, -0.001], $t(67) = -2.696$, $p = 0.009$, partial-$R^2 = 0.086$), with stronger effects for pre-to-in versus post-to-in transitions, indicating incomplete recovery post-flight. Movement direction showed no significant effect (all $p > 0.42$), suggesting that the relationship between corrective movements and duration slowing generalizes across reaching directions.

The analysis of primary submovements revealed even stronger evidence for feedforward control changes in microgravity. By focusing on just the primary submovement in two-peak trials, we found more pronounced effects than when analyzing the overall movement (comparing **Figure 5E** to **Figure S4C**). The primary submovement showed a robust reduction in peak speed during spaceflight (main effect of phase: $F(2,22) = 10.363$, $p = 0.001$, partial $\eta^2 = 0.485$) that was consistent across directions (interaction: $F(4,44) = 0.382$, $p = 0.711$; **Figure 5E**). More tellingly, while the overall movement showed timing changes only in relative measures, the primary submovement exhibited earlier peak speed timing in absolute terms (phase-by-direction interaction: $F(2,22) = 4.264$, $p = 0.012$, partial $\eta^2 = 0.279$). This temporal advance was most evident in directions with higher effective mass (planned contrasts vs. post-flight: 135°: $p = 0.008$, $\eta^2 = 0.615$; 90°: $p = 0.076$, $\eta^2 = 0.403$) but absent in the 45° direction ($p = 0.188$, $\eta^2 = 0.284$). The selective nature of the primary submovement in both magnitude and timing (comparing **Figure 4B** and **Figure 5F**) provides compelling evidence for mass-dependent changes in feedforward control. The control group showed only the expected directional variations ($F(2,22) = 46.259$, $p < 0.001$, partial $\eta^2 = 0.808$; **Figure S4D**) without any phase effects (all $p > 0.13$), confirming these changes as specific to microgravity exposure.

Figure 5.

Submovements analysis revealed changes in corrective movements during spaceflight.

A) Decomposition of a representative two-peak speed profile, illustrating primary and corrective submovements separated by the inter-peak interval (IPI). **B)** Proportion of movements showing corrective submovements across phases and directions. **C)** Magnitude of feedback corrections quantified by IPI, showing direction-dependent increases in microgravity. **D)** Linear relationship between feedback correction changes (ΔIPI) and movement slowing (ΔMD). Each participant (shown in different colors) contributed multiple data points from different phases and directions. **E-F)** Primary submovement characteristics: peak speed amplitude and timing. Error bars denote standard error across participants; asterisks indicate significance (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$).

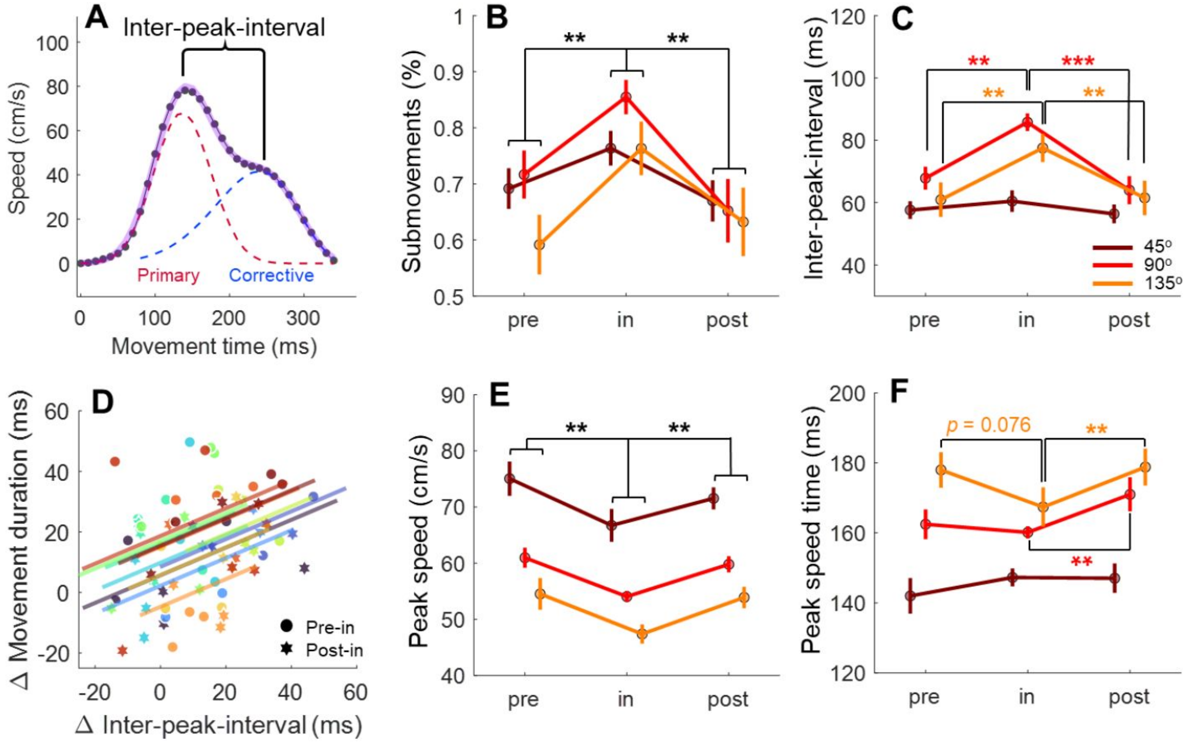
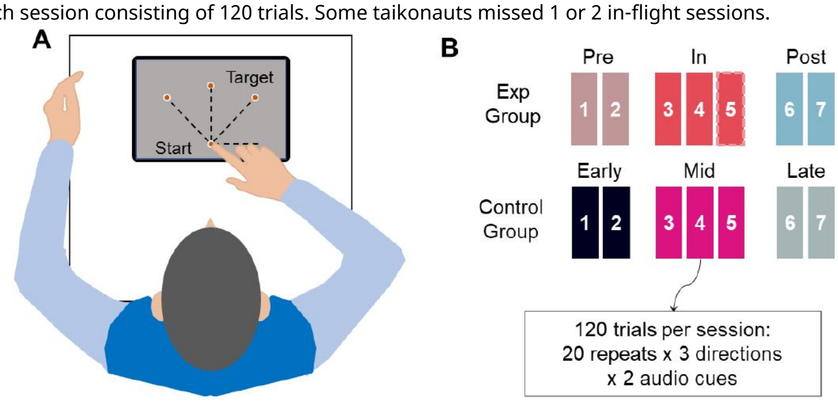


Figure 6.

Experimental Setup and Design.

A) Top-down view of a participant performing the reaching task with the right hand on a tablet. The start position and all possible target locations are shown as orange dots on the tablet screen. **B)** Experimental design. Both groups completed 4–7 sessions, with each session consisting of 120 trials. Some taikonauts missed 1 or 2 in-flight sessions.



Speed-accuracy trade-offs remain intact during spaceflight

The speed-accuracy trade-off, a fundamental indicator of motor control efficiency, remained largely intact during spaceflight. We evaluated this trade-off using two complementary approaches. First, following the traditional Fitts' framework (Plamondon & Alimi, 1997), we examined the relationship between movement duration and endpoint dispersion, which was not negatively affected by microgravity (**Figure S5**). Second, we analyzed the trade-off between reaction time and initial movement variance (**Figure S6**), which reflects the quality of movement planning (Sutter et al., 2021). This relationship persisted across all phases in both groups. Thus, while microgravity altered movement execution through mass underestimation, it did not fundamentally disrupt the sensorimotor system's capacity to regulate the speed-accuracy trade-off.

Discussion

Long-term exposure to microgravity presents a unique environment that cannot be replicated on Earth, offering invaluable insights into human sensorimotor adaptation. Understanding how microgravity affects motor performance is crucial both for ensuring successful space exploration and for advancing our fundamental knowledge of motor control principles. Our study elucidates the mechanisms underlying one of the most distinctive motor signatures in microgravity: the slowing of goal-directed actions during spaceflight. While taikonauts maintained reaching accuracy and rapid reaction times, their movements exhibited systematic changes in microgravity characterized by reduced peak speed and acceleration, along with advanced timing of these peaks. These kinematic alterations persisted across repetitive trials within a session and also persisted across multiple sessions during spaceflight (**Figure S7**), contradicting the hypothesis that astronauts adopt a conservative control strategy to maintain stability. Instead, they align precisely with our hypothesis that individuals underestimate limb effective mass, leading to underactuated initial movements. These changes followed model predictions across movement directions with varying effective masses due to limb biomechanics, confirming that the representation of body mass indeed affects movement control during spaceflight. Beyond the initial underactuation caused by altered feedforward control, we observed increased feedback-based corrective submovements that predicted movement duration prolongation in spaceflight. These effects were specific to microgravity exposure, occurring only during spaceflight with marked recovery post-flight, while ground controls showed no comparable changes. Notably, the taikonauts' fundamental motor control capabilities remained largely intact, as evidenced by preserved movement accuracy, reaction time, and speed-accuracy trade-offs.

The underactuation observed in initial movements under microgravity extends findings from parabolic flight studies (Crevecoeur et al., 2014; Papaxanthis et al., 2005). When individuals performed reaching movements with handheld weights during parabolic flight-induced gravity alterations, their reaching trajectories and grip forces showed systematic changes suggesting mass misperception (Bock, 1998; Bock et al., 1996a, 1996b; Crevecoeur et al., 2014; Papaxanthis et al., 2005). In zero-gravity conditions, goal-directed arm movements were slower (Papaxanthis et al., 2005), and grip-load force coordination was altered (Crevecoeur et al., 2010, 2014), consistent with mass underestimation. Conversely, hypergravity (1.8g) induced increased peak acceleration and speed in reaching movements, along with elevated grip force during object manipulation, suggesting mass overestimation (Bock, 1998; Bock et al., 1992, 1996a, 1996b). Notably, modified grip-load force coordination reflects changes in feedforward control (Flanagan & Wing, 1997; Johansson & Edin, 1993), indicating that mass misestimation influences action planning. However, while parabolic flight studies capture only transient adaptations to rapid gravity transitions, our study targets general slowing during spaceflight and reveals the enduring effects of sustained microgravity exposure.

Our results strongly challenge the conservative control strategy hypothesis, which posits that movement slowing reflects a generalized adaptation to address stability or safety concerns during spaceflight (Bock, 1998). Contrary to this hypothesis, taikonauts demonstrated faster reaction times during spaceflight—incompatible with a generalized slowing strategy. More critically, if the sensorimotor system adopted a conservative strategy with pre-planned longer movement durations, the speed profile should maintain symmetry with delayed peak acceleration and speed. Instead, we observed left-skewed speed profiles with earlier peak occurrence, directly contradicting the predictions of strategic slowing. This temporal change was more pronounced if primary submovement is isolated to pinpoint the feedforward control of the reaching since the overall speed profile is affected by the later feedback-based corrections. Previous studies reported inconsistent findings regarding speed profile symmetry, including advanced (Sangals et al., 1999), delayed (Fowler et al., 2008), or unchanged (Berger et al., 1997; Mechtcheriakov et al., 2002) peak speed timing during spaceflight. These discrepancies likely reflect methodological differences, including task variations and technical limitations. For example, two studies used joystick-based pointing tasks without involving limb movement (Fowler et al., 2008; Sangals et al., 1999), which arguably makes them less suitable for examining mass underestimation effects. Mechtcheriakov and colleagues used an outstretched arm task, but their limited sampling rate of 25 Hz may have hindered precise quantification of peak speed, preventing submovement analysis. Additionally, our larger sample size (n = 12) may have enhanced detection of microgravity effects compared to previous studies with typically 3-7 participants (Berger et al., 1997; Bock et al., 2001, 2003, 2010; Fowler et al., 2008; Mechtcheriakov et al., 2002; Papaxanthis et al., 1998; Weber & Stelzer, 2022) or single-participant designs (Heuer et al., 2003; Manzey et al., 1998; Sangals et al., 1999).

The underestimation of body mass in microgravity likely arises through two complementary mechanisms: degraded proprioceptive feedback and misinterpretation of weight-related cues. Microgravity impairs proprioception by reducing both muscle spindle sensitivity (Lackner & DiZio, 1992) and joint receptor responsiveness (Proske & Weber, 2023)—inputs that are crucial for weight perception (Proske & Gandevia, 2012). Furthermore, the sensorimotor system may interpret weight unloading as reduced body mass rather than environmental change (Bock et al., 1996b). Recent studies demonstrate that the sensorimotor system probabilistically infers body-versus-environment causality based on available sensory cues (Berniker & Kording, 2008; Fercho & Baugh, 2014; Kluzik et al., 2008; Kong et al., 2017; Wilke et al., 2013). In normal gravity, consistent mass-weight relationships lead to heavy reliance on weight-related cues for mass estimation (Bock, 1998; Bock et al., 1996b; Sangals et al., 1999). In microgravity, however, reduced weight cues become misleading and may systematically bias mass perception, affecting feedforward control of movement.

The persistence of deviant feedforward control and the subsequent within-movement corrections appears intriguing from the theories of human motor learning. Humans demonstrate remarkable ability to adapt to environmental changes in terrestrial studies. For experimentally-applied visual or force perturbations, individuals can recalibrate their sensorimotor control and restore baseline performance within minutes (Krakauer et al., 2019), a process often theorized as forming internal models of the new environment (Kawato, 1999; Wolpert et al., 1995). Relatedly, even when the body mass is perturbed by attaching a weight on the forearm, individuals can fully adapt their reaching movements within tens of trials (Wang & Sainburg, 2004). Our in-flight sessions were scheduled at least three weeks post-launch—even well beyond the 2-3 weeks window typically considered critical for sensorimotor adaptation in spaceflight (Kanas & Manzey, 2008), but the participants still exhibited un-adapted feedforward control and internal model.

We believe this contrast stems from distinct natures of perturbation effects in microgravity when compared to in terrestrial environments. Previous studies on sensorimotor adaptation simulated environmental changes by imposing a perturbation to alter the relationship between motor commands and sensory consequences of motor commands. For reaching adaptation here, visuomotor perturbations changes the spatial mapping between hand motion and its visual representation's motion (Krakauer et al., 2000; Wolpert et al., 2011); force perturbations change the dynamic mapping between actual force outputs and hand motion (Lackner & Dizio, 1994; Shadmehr & Mussa-Ivaldi, 1994). These mapping changes, though novel, are consistent thus the sensorimotor system is capable of approximating them and adapting accordingly. Theories of sensorimotor adaptation conceptualize the formation of the new mapping as acquiring internal models, which links motor command and efference copy to sensory consequences of motor commands (Flanagan & Wing, 1997; Kurtzer et al., 2008; Wolpert et al., 1998; Wolpert & Miall, 1996). However, the novel environment of microgravity alters the sensory estimate of motor apparatus (i.e., the body mass here), not merely the sensorimotor mapping. In normal gravity, weight-related sensory cues—mostly proprioceptive feedback—accurately represent body mass. In microgravity, however, these weight-related cues are substantially reduced due to the absence of gravitational pull, providing persistently biased information about body mass. Body mass is a parametric input to the internal models, thus its bias would affect motor actions even when the internal models are well adapted. Hence, the unique environment of microgravity reveals a fundamental constraint of the sensorimotor system, i.e., its quick adaptability is restricted to learning a novel sensorimotor mapping, not to persistent sensory bias of bodily property.

The discrete nature of our reaching task may contribute to the persistence of sensory bias. Within-movement corrections indicates that the sensorimotor system learns about sensory bias of mass estimation (Dimitriou et al., 2013), possibly via kinesthetic feedback during movement (Papaxanthis et al., 2005; Proske & Weber, 2023; Weber & Stelzer, 2022). However, movement-related cues only become available after movement initiation. Once a movement ends and the hand returns to rest, microgravity continues to elicit misleading sensory cues that bias mass estimation between trials. This lack of between-trial adaptation parallels findings from a manual interception study aboard the International Space Station, where astronauts persistently initiated movements too early when attempting to intercept an object moving at constant speed (McIntyre et al., 2001). Although astronauts corrected their movements online within each trial, their initial timing error persisted across multiple trials and sessions during a 15-day spaceflight. Hence, discrete motor tasks, such as manual interception and reaching, rely heavily on feedforward control, and are susceptible to sensory biases induced by microgravity. This explains why continuous reaching between targets, without stopping, shows quicker adaptation to zero gravity in parabolic flights (Papaxanthis et al., 2005) and does not exhibit prolonged movement duration during spaceflight (Fowler et al., 2008).

What are the potential consequences of body mass underestimation beyond the simple reaching movements examined here? The implications may be particularly significant given two key considerations. First, the majority of human movements are not deliberately controlled (Brooks, 1986; Marsden et al., 1976), suggesting that underactuated initial movements may be pervasive in microgravity—especially for actions involving larger body masses such as whole-body movements or object manipulation. Without the controlled conditions and explicit speed requirements of our experimental task, such underactuation might manifest even more prominently in natural movements during spaceflight. Second, our finding of increased reliance on feedback-based corrections points to a potentially costly compensatory mechanism. As feedforward control becomes systematically biased by mass underestimation, the sensorimotor system must depend more heavily on feedback control to achieve adequate performance. This increased reliance on feedback processes may demand additional cognitive resources for action regulation (Bock et al., 2010), potentially contributing to the non-specific stressors that have been shown to impair human performance in space (Tian et al., 2024). The cognitive cost of these compensatory adjustments might be particularly relevant for complex motor tasks or situations requiring divided attention during spaceflight operations.

The preservation of speed-accuracy trade-offs in our study extends previous findings from Fitts' task experiments performed during spaceflight (Fowler et al., 2008). Both the traditional speed-accuracy trade-off and action planning-dependent trade-off remained intact in microgravity (**Figure S6**), suggesting preserved fundamental motor control capacity. Motor performance during spaceflight typically degrades in tasks with high cognitive demands (Eddy et al., 1998; Manzey et al., 1998; Strangman et al., 2014), particularly those requiring sustained attention (Tian et al., 2024). Our findings suggest that while mass underestimation affects movement execution, the underlying motor control capabilities remain robust in microgravity when cognitive demands are modest.

Our study presents three methodological limitations that warrant consideration. First, despite matching for age and gender, our ground controls exhibited faster movements than the taikonauts. This systematic difference, reflected in higher peak acceleration and speed, may stem from differing experience with the touchpad device used for movement recording. While taikonauts used touchpads regularly, 9 of 12 control participants had minimal experience, potentially leading to compensatory faster movements to avoid overtime errors. Second, the temporal evolution of mass underestimation effects remains unclear. Our earliest measurements occurred three weeks post-launch, though parabolic flight studies suggest these effects may emerge within hours of microgravity exposure (Crevecoeur et al., 2014; Papaxanthis et al., 2005). Additionally, incomplete post-flight recovery in some kinematic measures suggests either a mass overestimation due to exaggerated weight-related cues or neuromuscular deficiency typically observed shortly after returning to earth (Tays et al., 2021). Third, while our hand-reaching task enabled us to use well-established analytical frameworks for studying motor control aspects of movements, the generalizability of mass underestimation effects to other movements, particularly whole-body actions, remains to be determined. Future research should address these limitations by implementing matched device training protocols, adding early- and post-flight measurements, and extending investigations to diverse motor tasks.

In conclusion, our study provides compelling evidence that movement slowing in microgravity stems from systematic underestimation of body mass, rather than a conservative control strategy. By leveraging the direction-dependent variation in limb effective mass, we demonstrated that reaching movements in microgravity exhibit kinematic signatures—reduced and earlier peak speed/acceleration—that precisely match the predictions of mass underestimation. The persistence of these effects throughout long-duration spaceflight, coupled with increased feedback-based corrections, reveals how altered sensory inputs fundamentally impact both predictive and reactive components of motor control and evades sensorimotor adaptation. While basic motor control capabilities remain intact, as evidenced by preserved speed-accuracy trade-offs, the consistent underactuation of movements may have significant implications for the theorization of sensorimotor learning and for practical considerations of motor performance during space operations. These findings not only advance our understanding of how the sensorimotor system adapts to novel gravitational environments but also reveal fundamental principles of sensorimotor integration, particularly the central role of gravitational cues in shaping our perception of body dynamics.

Methods

Participants

Twelve taikonauts (2 females, 10 males; mean age 49.5 ± 6.5 years) from the first four missions of the China Space Station (CSS) served as the experimental group (Table S1). The control group consisted of 12 right-handed residents from Beijing (2 females, 10 males; mean age 49.9 ± 6.5 years). Control group received monetary compensation. All participants were provided with written informed consent forms, approved by the Ethics Committee of the China Astronaut

Research and Training Center and the Institutional Review Board of Peking University. The experimental group was exposed to microgravity for 92–187 days during manned missions in China's Shenzhou Program. Due to operational constraints, the number of test sessions varied among taikonauts: one completed 4 sessions, five completed 6, and six completed 7 (Table S1). All control sessions were conducted at Peking University.

Task and experimental design

The task was conducted using a tablet (Surface Pro 6, Microsoft, Redmond, Washington) and noise-canceling earphones (FiiO FA7, FiiO Electronics Technology, Guangzhou, China). During in-flight sessions, participants wore earphones and assumed a neutral posture with feet secured by foot straps in front of a foldable tabletop, positioned 0.9 meters above the cabin floor. They held the tabletop edge with their left hand, allowing their right hand to move freely on the tablet (**Figure 6A**). The foot straps and left-hand grip provided body stabilization. The tablet, attached to the tabletop with Velcro, was placed ~35 cm in front of participants and 30 cm below chest level (**Figure 6A**). At the start of each trial, an orange circle (0.5 cm radius) appeared at the bottom of the screen, signaling participants to move their right index finger to this start circle. After a random delay (500–1000 ms), a target appeared at one of three possible locations, 12 cm from the start circle at 45°, 90°, or 135° counterclockwise from the horizontal axis (**Figure 6A**). Participants were instructed to reach the target quickly and accurately, pausing briefly before returning to the start. If the time from target appearance to movement end (finger speed <10 cm/s) exceeded 650 ms, a "too slow" message prompted faster movement. In half of the trials, a 50 ms "beep" sound was played at target appearance. Each session included 120 trials, with 3 target directions and 2 beep conditions balanced and randomized (**Figure 6B**). The x-y position of the index finger was recorded continuously at 100 Hz.

The control group was included to assess a potential confounding effect of repeated measurements. The interval between successive test sessions was 6–9 days, which was shorter than those of the taikonauts (1–2 months) to conservatively evaluate potential practice effects. Despite the shorter intervals, no significant differences were observed across sessions for most measures (except reaction time), confirming minimal practice effects and providing a robust baseline for evaluating microgravity's impact on the taikonauts. All ground tests, for both groups, were conducted using identical devices and software. The layout of the table and tablet was kept consistent across sessions. The only difference was that, in ground tests, participants sat in a chair approximately 90 cm above the ground without foot straps and did not need to hold the edge of the tabletop with their left hand for stability.

Data analysis

Kinematic analysis

A Butterworth low-pass filter with a 20 Hz cutoff frequency was applied to the positional data. Movement distance at each time point was computed as the Euclidean distance between the finger's current position and the center of the start circle. Speed was estimated by performing linear regression over a moving window of five consecutive positional data points, and acceleration was calculated similarly from the speed data. Movement onset was defined as the first time point at which acceleration exceeded 50 cm/s^2, while movement offset was the first time point at which acceleration fell below 50 cm/s^2. Reaction time was defined as the interval between target onset and movement onset, and movement duration as the interval between movement onset and offset. Movement endpoint error was computed as the Euclidean distance between the target and the finger's position at movement offset. Peak speed and peak acceleration were identified as the maximum speed and acceleration occurring between movement onset and offset, respectively. The times of peak speed and peak acceleration were measured as their respective time intervals from movement onset, and their relative times were calculated by dividing each by the movement duration.

Submovement extraction

To separate the primary submovement from the secondary corrective submovement, the speed profile of each reaching movement was decomposed into a sum of submovements. Following the optimization algorithm described by Rohrer and Hogan (2006), the speed profile was fitted as a sum of support-bounded lognormal (LGNB) curves. Each LGNB curve is defined as:

$$B(t) = \frac{(T_1 - T_0)D}{\sigma\sqrt{2\pi}(t - T_0)(T_1 - t)} exp\left\{\left(\frac{-1}{2\sigma^2}\right)\left[ln\left(\frac{t - T_0}{T_1 - t}\right) - \mu\right]^2\right\},$$

where T_0 and T_1 represent the start and end times of the movement, D is a scaling parameter, and the parameters μ and σ determine the skewness and kurtosis of the underlying lognormal function. These five parameters enable each LGNB submovement to vary across a wide range of possible shapes.

In theory, the number of submovements could be treated as a free parameter. However, to avoid overfitting, we limited each movement to two submovements, given the brevity of the movement (average duration approx. 350 ms). For each trial, we initially fit the data using both one and two LGNB curves and then determined the optimal number of submovements using the “greedy” method from the previous study (Rohrer & Hogan, 2006). Briefly, the fitting error for each movement with one and two LGNB curves was calculated as:

$$\varepsilon = \frac{\int |F(t) - G(t)| dt}{\int |G(t)| dt},$$

where G(t) is the actual movement speed profile, and F(t) is the fitted speed profile (the sum of the submovements). If the fitting error for a single submovement was below a 2% threshold, or if adding a second submovement reduced the fitting error by less than 2%, a single submovement was considered adequate. Otherwise, two submovements were included. Trials with large fitting errors (>10%) were discarded, accounting for 3.60% and 4.83% of trials in the experimental and control groups, respectively. The percentage of trials with two submovements was calculated for each movement direction and phase. The time difference between the peaks of the two submovements was also computed, and set to zero if only one submovement was detected. Data fitting was performed using Matlab's *fmincon* function (MathWorks, Natick, MA, USA), and the robustness of fitting was confirmed by testing 20 random sets of initial parameters.

Statistical analysis

First, invalid trials were excluded from further analysis, with three possible causes: early initiation (RT < 150 ms, indicating guessing of the target direction), affecting 0.74% and 0.70% of trials in the experimental and control groups, respectively; late initiation (RT > 400 ms, indicating inattention), affecting 1.13% and 0%; and measurement failures, affecting 1.16% and 0.70%. The proportion of invalid trials did not differ significantly across testing phases (Friedman's tests, all $p > 0.1$).

Most dependent measures—including movement duration, peak acceleration, peak speed, time to peak acceleration/speed, relative time to peak acceleration/speed, the percentage of submovements, and inter-peak intervals—were analyzed using two-way repeated-measures ANOVAs with a 3 (phase) × 3 (movement direction) design. Reaction time was analyzed using three-way repeated-measures ANOVAs with a 3 (phase) × 3 (movement direction) × 2 (beep/no beep) design. Greenhouse-Geisser corrections were applied when sphericity was violated (Kirk, 1968). Normality was assessed with Shapiro-Wilk tests, and if more than 10% of tests for a dependent variable violated normality, the Aligned Rank Transform (ART) procedure was applied before conducting repeated-measures ANOVA. Tukey's HSD tests were used for pairwise comparisons.

To examine the relationship between submovement and movement duration, a Linear Mixed Model (LMM) was used to analyze the association between changes in the inter-peak interval of submovements (ΔIPI) and changes in movement duration (ΔMD). The dependent variable $y_{\Delta MD}$represents the mean change in movement duration across two adjacent phases (pre-to-in and post-to-in transitions) for each target direction. Target direction (45°, 90°, 135°) and phase transitions (pre-to-in and post-to-in) were treated as within-subject factors. The predictors included three fixed factors: ΔIPI, phase transition (x_{pha}), and direction (x_{dir}). An intercept term (u_{par}) was included as a random factor to account for individual differences in movement duration. The resulting LMM is as follows:

$$y_{\Delta MD} = \beta_0 + \beta_1 x_{\Delta IPI} + \beta_2 x_{pha} + \beta_3 x_{dir} + Z u_{par} + \varepsilon,$$

where β_0 denotes the overall intercept and ε denotes a Gaussian residual error.

All statistical analyses were performed using MATLAB's Statistics and Machine Learning Toolbox. The significance level was set at α = 0.05, and all tests were two-tailed unless otherwise specified.

Data and materials availability

Experimental data of the control group will be made available upon reasonable request.

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Additional information

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Authors contributions

Conceptualization: KW, YT, ZZ; Data curation: YT, ZZ; Formal analysis: ZZ; Funding acquisition: KW, CW, YT, ZZ; Methodology: KW, ZZ; Software: YT, CW, CJ, BW, HY, RZ; Supervision: KW; Visualization: ZZ; Writing: KW, ZZ.

Supplemental Materials

Supplemental figures and tables

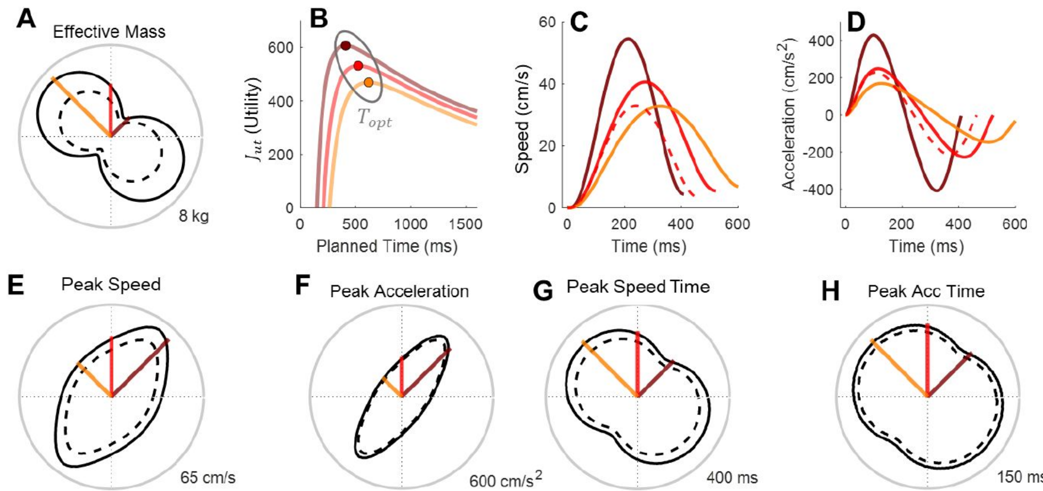

Figure S1.

Simulation details for the arm model.

A) Effective mass across different movement directions, with three colored lines indicating the specific directions used in the experiment. The solid-line contour denotes the effective mass simulated by default parameters under normal gravity (see Supplemental Text 1), and the dash-line contour denotes a hypothetical 30% underestimation of the effective mass in microgravity **B)** Utility function values for movements in each of the three directions. **C-D)** Speed and acceleration profiles for movements in these directions. **E-H)** Simulated values of peak speed, peak acceleration, peak speed timing, and peak acceleration timing across all movement directions. The solid-line contours denote simulated values by default parameters, and the dash-line contours denotes the values with a 30% mass underestimation.

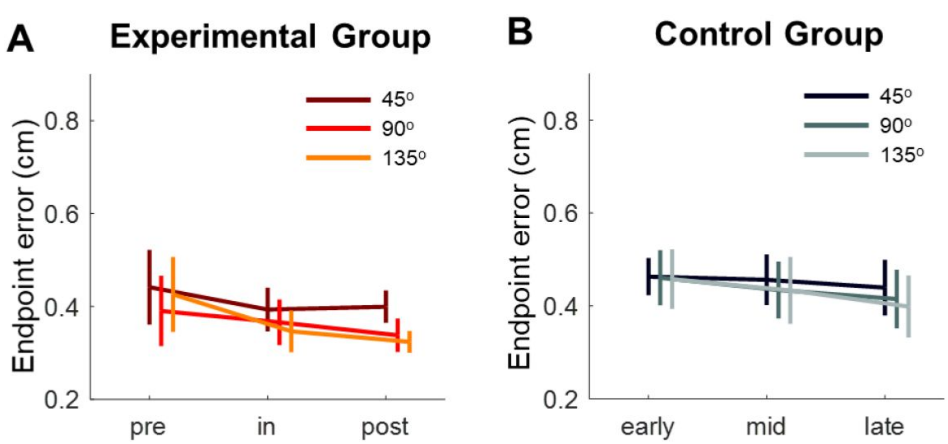

Figure S2.

Movement accuracy quantified by endpoint error.

The average endpoint error for each target was analyzed across experimental phases. Both the experimental group **A)** and control group **B)** performed the task with high accuracy despite the stringent timing requirements. The presence of a beep had no effect on endpoint error in either group (experimental group: main effect, F(1,11) = 1.4461, p = 0.254; all interactions, p > 0.2; control group: main effect, F(1,11) = 0.717, p = 0.415; all interactions, p > 0.25). Therefore, data from the two beep conditions were pooled, and a two-way repeated-measures ANOVA was conducted for each group. In the experimental group, endpoint error showed a slight decrease across phases, but the main effect of phase was not statistically significant (F(2,22) = 1.768, p = 0.210), nor was the interaction effect (F(2,22) = 2.386, p = 0.091). However, there was a significant main effect of direction (F(2,22) = 16.020, p < 0.001, partial $\eta^2 = 0.593$), with post-hoc comparisons revealing that the 45° direction had significantly larger endpoint errors than the 90° and 135° directions (p = 0.002, partial $\eta^2 = 0.291$; p = 0.001, partial $\eta^2 =$ 0.286, respectively). In the control group, no significant effects of phase or direction were observed, and there was no significant interaction (all p > 0.23).

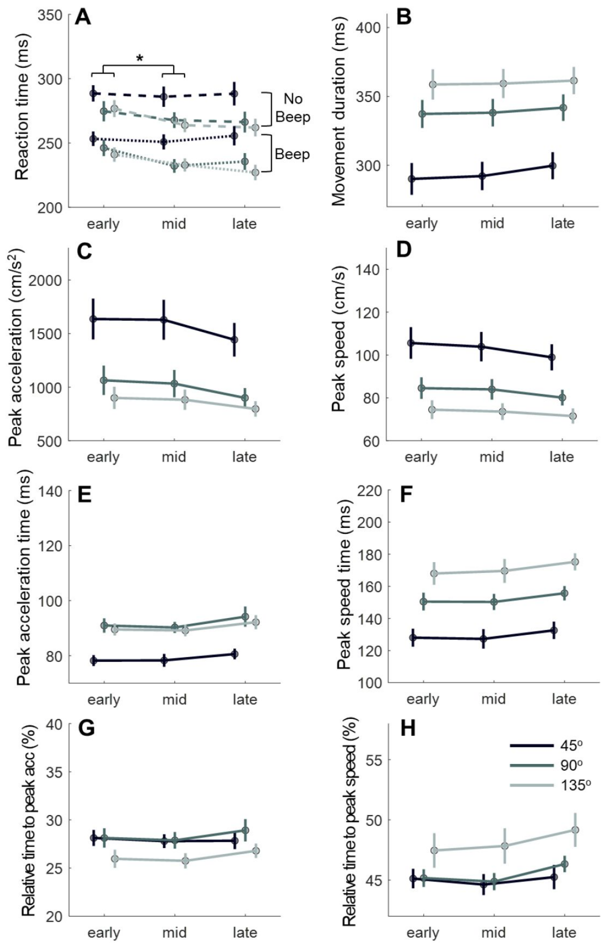

Figure S3.

Results of kinematic analysis of the control group.

The control group exhibited no phase effect for most performance measures, with the exception of reaction time (RT). However, similar effects were also observed in the experimental group. Each measure was analyzed using a 3 (direction) × 3 (phase) two-way repeated-measures ANOVA. **A)** Reaction Time (RT): Significant main effects of direction ($F(2,22) = 21.946$, $p < 0.001$, partial $\eta^2 = 0.666$) and phase ($F(2,22) = 4.281$, $p = 0.039$, partial $\eta^2 = 0.280$) were found, along with a significant direction-phase interaction ($F(2,22) = 7.940$, $p = 0.001$, partial $\eta^2 = 0.419$). RT was significantly faster in the Middle phase than in the Early phase ($p = 0.032$, D = 0.285). **B)** Movement Duration (MD): Significant main effect of direction ($F(2,22) = 99.874$, $p < 0.001$, partial $\eta^2 = 0.901$), with all pairwise comparisons significant at $p < 0.001$. Neither main effect of phase ($F(2,22) = 0.995$, $p = 0.360$) nor interaction effect ($F(4,44) = 1.412$, $p = 0.114$) was significant. **C)** Peak acceleration: Significant main effect of direction ($F(2,22) = 55.460$, $p < 0.001$, partial $\eta^2 = 0.834$), with all pairwise comparisons significant at p < 0.001. A marginal phase effect ($F(2,22) = 3.462$, $p = 0.081$) and interaction ($F(4,44) = 2.715$, $p = 0.067$) were noted. **D)** Peak speed: Significant main effect of direction ($F(2,22) = 66.071$, $p < 0.001$, partial $\eta^2 = 0.857$), with all pairwise comparisons significant at p < 0.001. No phase effect ($F(2,22) = 2.360$, $p = 0.148$) or interaction ($F(4,44) = 1.975$, $p = 0.145$) was observed. **E)** Peak acceleration time: Significant main effect of direction ($F(2,22) = 26.602$, $p < 0.001$, partial $\eta^2 = 0.707$). No phase effect ($F(2,22) = 1.440$, $p = 0.260$) or interaction ($F(4,44) = 0.132$, $p = 0.900$) was observed. Pairwise comparisons showed significantly shorter times for 45° compared to 90° ($p < 0.001$, D = 0.305) and 135° ($p < 0.001$, D = 0.269). **F)** Peak speed time: Significant main effect of direction ($F(2,22) = 96.575$, $p < 0.001$, partial $\eta^2 = 0.898$), with all pairwise comparisons significant at $p < 0.001$. Marginal phase effect ($F(2,22) = 2.702$, $p = 0.098$), and no interaction effect ($F(4,44) = 0.420$, $p = 0.710$). **G)** Relative peak acceleration time: Significant main effect of direction ($F(2,22) = 8.506$, $p = 0.003$, partial $\eta^2 = 0.436$), with no phase effect ($F(2,22) = 1.143$, $p = 0.317$) or interaction ($F(4,44) = 1.002$, p = 0.381). Pairwise comparisons indicated that relative peak acceleration time appeared significantly earlier in the 135° direction than in 90° ($p = 0.002$, D = 0.313) and 45° ($p = 0.021$, D = 0.256). **H)** Relative peak speed time: Significant main effect of direction ($F(2,22) = 6.779$, $p = 0.007$, partial $\eta^2 = 0.381$), with a marginal phase effect ($F(2,22) = 2.636$, $p = 0.095$) and no interaction ($F(4,44) = 1.788$, $p = 0.175$). Pairwise comparisons showed that relative peak speed time was significantly earlier in the 135° direction than in 90° ($p = 0.037$, D = 0.264) and 45° ($p = 0.029$, D = 0.308).

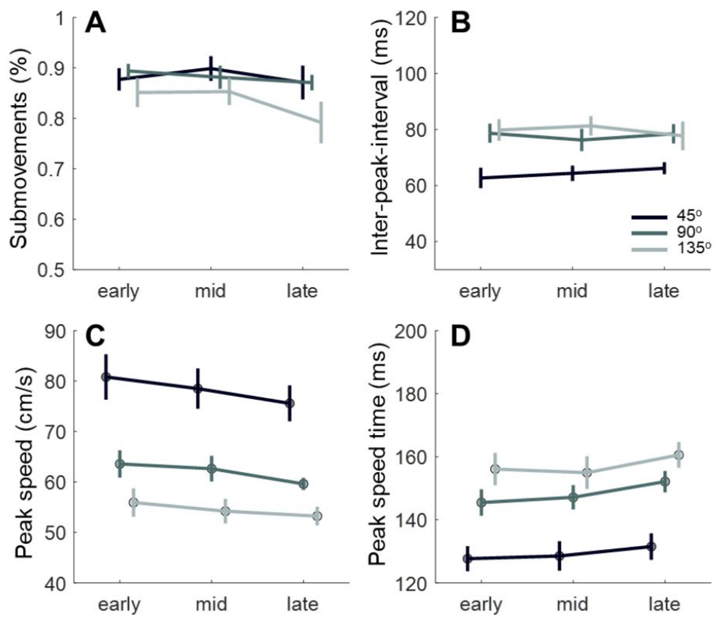

Figure S4.

Results of submovement analysis of the control group.

All measures of submovements did not show significant phase effect. **A)** Percentage of trials with two submovements. Only a marginal main effect of direction was detected (F(2,22) = 3.358, p = 0.055, partial η^2 = 0.234) without a phase effect (F(2,22) = 2.216, p = 0.155) or interaction (F(4,44) = 1.209, p = 0.320). **B)** Inter-peak interval (IPI) showed a significant main effect of direction (F(2,22) = 19.277, p < 0.001, partial η^2 = 0.637), with no phase effect (F(2,22) = 0.029, p = 0.902) or interaction (F(4,44) = 2.028, p = 0.134). Pairwise comparisons indicated a significantly shorter IPI in the 45° direction compared to the other two directions (45° vs. 90°: p < 0.001, D = 0.269; 45° vs. 135°: p = 0.002, D = 0.305). **C)** Peak speed of the primary submovement exhibited a significant main effect of direction (F(2,22) = 60.089, p < 0.001, partial η^2 = 0.845), without a phase effect (F(2,22) = 2.628, p = 0.120) or interaction (F(4,44) = 0.949, p = 0.420). Pairwise comparisons showed significant differences between all directions (45° vs. 90°: p < 0.001, D = 0.224; 45° vs. 135°: p < 0.001, D = 0.326; 90° vs. 135°: p < 0.001, D = 0.102). **D)** Peak speed time of the primary submovement revealed a significant main effect of direction (F(2,22) = 46.259, p < 0.001, partial η^2 = 0.808), with no phase effect (F(2,22) = 2.335, p = 0.134) or interaction (F(4,44) = 0.425, p = 0.660). Pairwise comparisons showed a significantly smaller peak speed time in the 45° direction compared to the other two directions (45° vs. 90°: p < 0.001, D = 0.260; 45° vs. 135°: p < 0.001, D = 0.311).

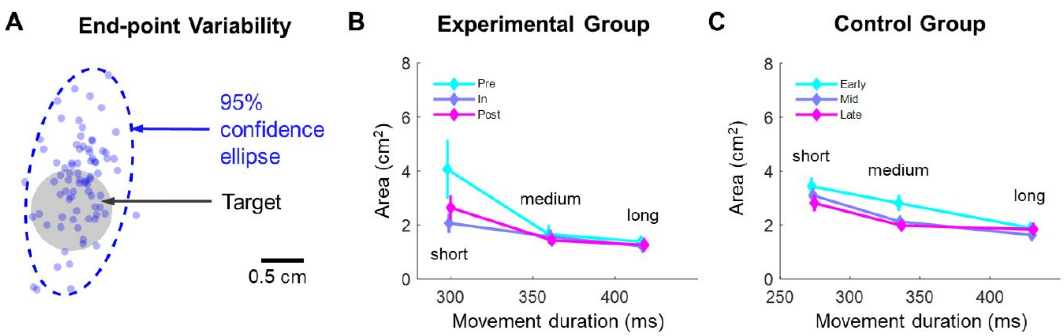

Figure S5.

Speed-accuracy trade-off as indicated by the relationship between movement duration (MD) and endpoint dispersion, following the classical Fitts' law.

A) Endpoint dispersion was estimated by constructing a 95% confidence ellipse encompassing the endpoints of all trials. Regardless of movement direction, phase, and beep condition, trials were divided into three equal-sized bins based on duration, labeled as short, medium, and long. For each participant at each phase, trials within each bin were pooled to estimate the ellipse of endpoints. **B-C)** A positive association between endpoint area and MD was observed, indicating a speed-accuracy trade-off. Both groups exhibited a dependency of the endpoint area on MD. For the experimental group, a 3 (phase) × 3 (MD) two-way repeated-measures ANOVA on endpoint area showed a significant main effect of MD ($F(2,22) = 14.125$, $p = 0.003$, partial $\eta^2 = 0.407$), with only a marginal phase effect and no interaction (phase: $F(2,22) = 3.199$, $p = 0.087$; interaction: $F(4,44) = 2.106$, $p = 0.163$). Pairwise comparisons indicated that short, medium, and long MD conditions significantly differed from each other (short vs. medium: $p = 0.018$, D = 0.576; medium vs. long: $p = 0.007$, D = 0.113; short vs. long: $p = 0.004$, D = 0.689). The control group yielded similar ANOVA results, with a significant main effect of MD ($F(2,22) = 34.058$, $p < 0.001$, partial $\eta^2 = 0.756$; all post-hoc comparisons were significant at $p < 0.007$). Additionally, the control group exhibited a main effect of phase ($F(2,22) = 9.359$, $p = 0.003$, partial $\eta^2 = 0.460$) without interaction ($F(2,22) = 2.026$, $p = 0.151$, partial $\eta^2 = 0.156$). Post-hoc comparisons revealed that the early phase had a significantly larger area than the middle and late phases (early vs. middle: $p = 0.004$, D = 0.263; early vs. late: $p = 0.018$, D = 0.309). Thus, both groups appear to slightly improve their speed-accuracy trade-off over repeated tests, with no evidence suggesting that spaceflight adversely affects this trade-off.

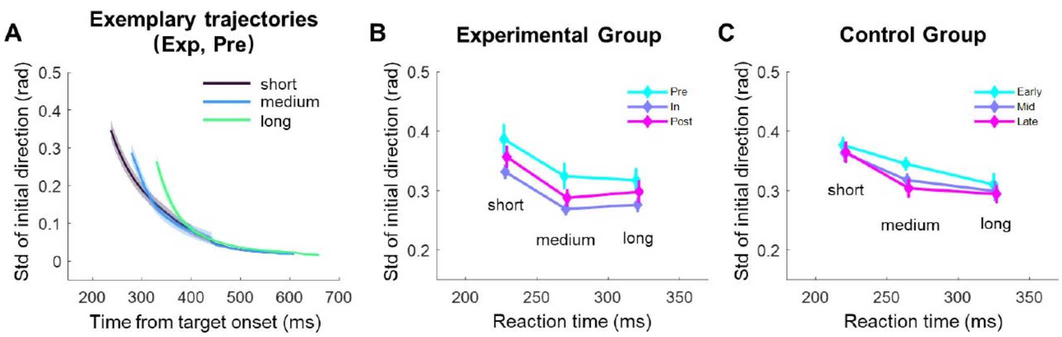

Figure S6.

Speed-accuracy trade-off in action planning, illustrated by the relationship between reaction time (RT) and the variance in initial movement direction.

A) Similar to the speed-accuracy trade-off analysis in **Figure S5**, all trials were grouped into three equal-sized bins based on RT, regardless of phase, movement direction, or beep condition. The standard deviation of movement direction is shown over time for these bins, using data from the experimental group in the pre-flight phase as an example. The standard deviation decreases over time within a reaching movement, with shorter RTs resulting in greater directional variance, particularly in the early phase of movement. **B-C)** The standard deviation of initial movement direction as a function of RT, plotted separately for the three phases. A negative trend between standard deviation and RT indicates a speed-accuracy trade-off in action planning. However, phase did not affect the relationship between RT and initial variance. For the experimental group, a 3 (phase) × 3 (RT) two-way repeated measures ANOVA on the standard deviation showed a significant main effect of RT ($F(2,22) = 7.542$, $p = 0.009$, partial $\eta^2 = 0.407$), while neither the main effect of phase nor the interaction was significant (phase: $F(2,22) = 3.160$, $p = 0.083$; interaction: $F(4,44) = 0.503$, $p = 0.694$). Pairwise comparisons indicated that the initial variance of short-RT trials was significantly higher than that of medium-RT trials ($p = 0.004$, D = 0.375), while the difference between medium- and long-RT trials was only marginally significant ($p = 0.076$). The control group showed similar results (RT: $F(2,22) = 4.268$, $p = 0.040$, partial $\eta^2 = 0.280$; phase: $F(2,22) = 1.657$, $p = 0.215$; interaction: $F(2,22) = 0.007$, $p = 0.998$). Thus, while clear indications of a speed-accuracy trade-off in action planning were observed, there was no evidence that microgravity negatively affected this relationship.

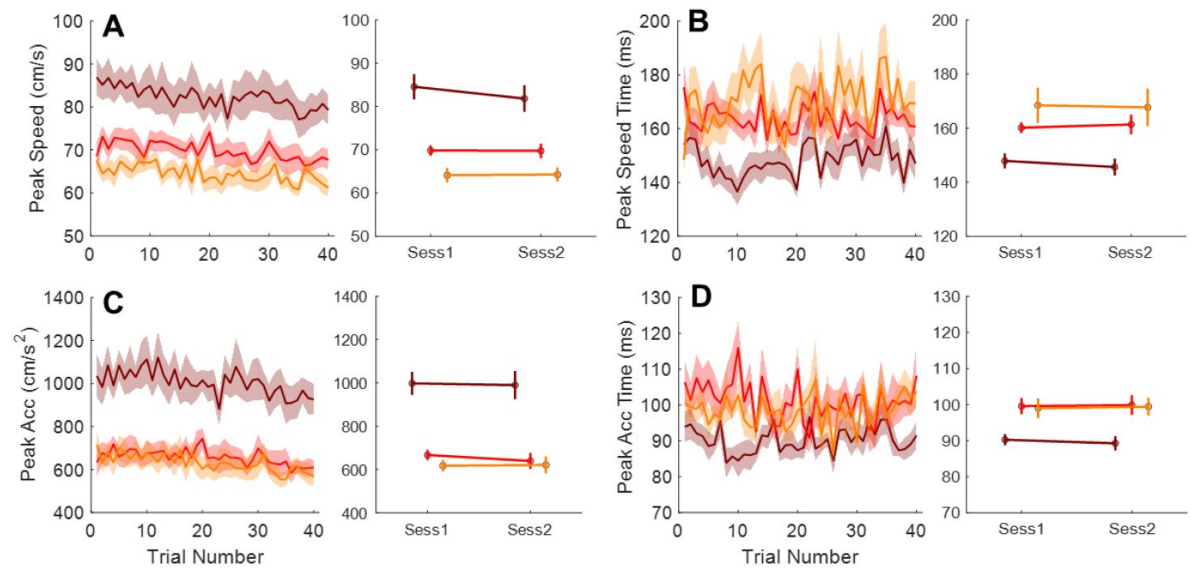

Figure S7.

Changes in key performance measures over trials within and between in-flight sessions.

To show a lack of within-session and between-session adaptation during spaceflight, we analyzed changes in peak speed **(A)**, peak speed timing of the primary submovement **(B)**, peak acceleration **(C)**, and peak acceleration timing **(D)** across the first and second sessions, as well as trials within the second in-flight session. The second session was specifically analyzed because the sensorimotor system undergoes recalibration during early flight and thus the first session is less likely to show good within-session adaptation. The third session had fewer participants (only six taikonauts). Participant S2, who completed only one in-flight session, was excluded from this analysis. For the within-session analysis, we compared the mean values of the first and last five trials for each measure across movement directions. No significant changes were observed in peak speed timing and peak acceleration timing (all *p*-values > 0.06; panels B and D). Peak speed (panel A) showed a slight and marginally significant decrease in the 45° and 90° directions (45°: from 85.9 ± 3.4 cm/s to 78.5 ± 3.4 cm/s, *p* = 0.013; 90°: from 71.6 ± 2.1 cm/s to 67.3 ± 1.8 cm/s, p = 0.052), but no significant change in the 135° direction (*p* = 0.200). Similarly, peak acceleration (panel C) decreased slightly across all directions with marginal significance (45°: from 1033.9 ± 73.4 cm/s^2 to 919.9 ± 66.8 cm/s^2, *p* = 0.056; 90°: from 671.8 ± 43.4 cm/s^2 to 603.7 ± 33.8 cm/s^2, *p* = 0.097; 135°: from 663.2 ± 46.3 cm/s^2 to 586.2 ± 36.2 cm/s^2, *p* = 0.049). For comparisons between the first and second sessions, we applied a two-way repeated measures ANOVA to these dependent variables and found no significant differences between sessions (all *p*-values > 0.144). These findings suggest that the mass underestimation effects persist over repetitive reaching attempts (120 trials) and practice sessions. Instead of showing between-trial or between-session adaptation, these effects tend to increase slightly across trials.

Taikonauts	Pre-flight		In-flight			Post-flight	
S1	-43	-24	39	74	--	14	66
S2	-43	-24	39	--	--	14	--
S3	-43	-24	39	74	--	14	66
S4	-157	-60	32	96	131	18	73
S5	-165	-60	32	96	131	18	73
S6	-165	-60	32	96	131	18	73
S7	-31	-12	20	109	--	15	64
S8	-31	-12	20	109	--	15	64
S9	-31	-12	20	109	--	15	64
S10	-91	-48	28	76	125	16	93
S11	-91	-50	28	77	125	16	93
S12	-91	-52	28	76	125	16	93

Numbers in the table represent days before launch (pre-flight), days after launch (in-flight), and days after landing (post-flight).

Table S1.

The schedule of experiment sessions.

Supplementary Note 1. Model simulation for illustrating the effect of effective mass on reaching kinematics

We simulated the peak speed, peak acceleration, and their corresponding times for reaching movements in different directions. We employed the classical two-joint arm model to demonstrate the effect of biomechanics ((Li & Todorov, 2004); **Figure 1B**). This was combined with a movement utility model to estimate planned movement time ((Shadmehr et al., 2016); **Figure S6A** and S5B) and a forward optimal controller to simulate reaching movements ((Todorov, 2005); **Figure 1C** and **F**, **Figure S6C-D**). Our simulation is briefly outlined below in steps.

Effective mass of the hand as a function of movement direction

We used a two-joint planar arm model to estimate the effective mass of the hand moving in different directions (**Figure 1B**). The parameters of the upper arm (length l_1, mass m_1, center of mass lc_1, inertia I_1) and the lower arm (length l_2, mass m_2, center of mass lc_2, inertia I_2) are based on values from Shadmehr et al., 2016:

$$l_1 = 0.33 \quad l_2 = 0.43 \qquad \text{m}$$

$$m_1 = 1.93 \quad m_2 = 1.52 \qquad \text{kg}$$

$$lc_1 = l_1/2 \quad lc_2 = 2l_2/3 \qquad \text{m}$$

$$I_1 = 0.014 \quad I_2 = 0.019 \qquad \text{kg.m}^2$$

The elbow and shoulder angles are denoted by θ_e and θ_s, respectively. The initial arm configuration is set by $\theta_s = 0.994$ rad, and $\theta_e = 1.5724$ rad, following the experimental conditions from Gordon et al., 1982. With a particular joint angle configuration, the end-point hand position is:

$$x = \left[l_1 \cos cos\, \theta_s + l_2 \cos cos\left(\theta_s + \theta_e\right);\ l_1 \sin sin\, \theta_s + l_2 \sin sin\left(\theta_s + \theta_e\right)\ \right]. \qquad \text{(eq.1)}$$

The arm's inertia matrix is:

$$I(\theta_s, \theta_e) = \left[I_{11}, I_{12}; I_{21}, I_{22}\right], \qquad \text{(eq.2)}$$

where:

$$I_{11} = a_3 + a_1 l_1^2 + a_4 + 2a_2 l_1 cos\theta_e,\ I_{12} = I_{12} = a_2 + l_1 cos\theta_e + a_4,\ I_{22} = a_4,$$

with:

$$a_1 = m_2,\ a_2 = m_2 lc_2,\ a_3 = m_1 lc_1^2 + i_1,\ a_4 = m_2 lc_2^2 + i_2$$

The joint torques at the shoulder and elbow are represented as:

$$\tau = I(\theta)\,\ddot{\theta}, \qquad \text{(eq.3)}$$

and the forces at the hand are:

$$f = M(\theta)\ddot{x}\,. \qquad \text{(eq.4)}$$

We use the Jacobian matrix $\Lambda = \frac{dx}{d\theta}$ to relate force and torque, based on virtual work principle:

$$\tau = \Lambda^T f, \quad \text{(eq.5)}$$

$$\dot{x} = \Lambda\dot{\theta}, \quad \text{(eq.6)}$$

$$\ddot{x} = \dot{\Lambda}\dot{\theta} + \Lambda\ddot{\theta}. \quad \text{(eq.7)}$$

Put S3 into S5:

$$f = \Lambda^{-1^T} I(\theta)\, \ddot{\theta}, \quad \text{(eq.8)}$$

then combine S7 and S8:

$$f = \Lambda^{-1^T} I(\theta)\Lambda^{-1}\left(\ddot{x} - \dot{\Lambda}\dot{\theta}\right). \quad \text{(eq.9)}$$

Since the hand speed at the beginning of the movement is zero, according to S4 and S9, the mass matrix of hand $M(\theta)$ is:

$$M(\theta) = \Lambda^{-1^T} I(\theta)\Lambda^{-1}, \quad \text{(eq.10)}$$

The mass matrix $M(\theta)$ is a 2 x 2 matrix, and the effective mass $m(\theta)$ is determined by the length of the force vector when $M(\theta)$ is subjected to a unit acceleration.

In **Figure S6A**, the solid curve illustrates the effective mass across various directions under normal gravity, while the dashed curve represents the effective mass with a hypothetical 30% ($m_{og}(\theta) = 0.\,3m(\theta)$)mass underestimation in microgravity. The colored lines indicate the effective mass amplitudes for the three target directions used in our experiment: 45°, 90°, and 135°.

Movement time determined by optimal utility

Following Shadmehr et al., 2016, the utility of an reaching action is given by:

$$J_{ut} = \frac{\alpha - amT - bmd^i/T^2}{1-\gamma T}, \quad \text{(eq.11)}$$

where α is the reward, m is mass, d is the distance to be moved, T is the planned duration of the movement, and a, b and i are scaling factors of the effort cost. γ represents a temporal discounting factor. The optimal movement time (T_{opt}) is the time that maximizes the utility J_{ut}:

$$T_{opt} = arg_T max\,(J_{ut}). \quad \text{(eq.12)}$$

The parameter values used in the simulation of **Figure S6B** were taken from Shadmehr 2016: $\alpha = 1000$, $a = 15$, $b = 100$, $i = 1.\,1$, $\gamma = 1$. The movement distance d is set as 0.12 m, the target distance in the experiment. **Figure S1B** illustrates how the utility curves change with planned movement time in three experimental directions (45°, 90°, and 135°) under normal gravity, with colored dots indicating the optimal movement durations.

Movement kinematics simulated by optimal control theory

After setting the movement time T_{opt}, the position, speed and acceleration over time can be simulated based on an optimal control model (Todorov, 2005). The arm's control operates as a second-order low-pass filter:

$$u_t = \tau_1\tau_2\ddot{f}_t + \left(\tau_1 - \tau_2\right)\dot{f}_t + f_t, \qquad \text{(eq.13)}$$

with the time constants $\tau_1 = \tau_2 = 40ms$. The system state vector is $x = \left[p;\ \dot{p};\ f;\ g;\ p^*\right]$,where p is position of hand, f is force acting on the hand, and g is an auxiliary state variable, p* is the target position. The control law is given by:

$$u_t = \tau_1\dot{g}_t + g_t,\ g_t = \tau_2\dot{f}_t + f_t.$$

The system dynamics are:

$$x_{t+1} = Ax_t + Bu_t,\ y_t = Hx_t, \qquad \text{(eq.14)}$$

$$A = \begin{bmatrix} 1 & dt & 0 & 0 & 0 \\ 0 & 1 & dt/m & 0 & 0 \\ 0 & 0 & 1 - dt/\tau_2 & dt/\tau_2 & 0 \\ 0 & 0 & 0 & 1 - dt/\tau_1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad H = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$B = \left[0;\ 0;\ 0;\ dt/\tau_1;\ 0\right].$$

The controller seeks to minimize the cost function J by optimizing the command u:

$$J = \int_0^\infty E\left[x^TQx + u^TRu\right]dt. \qquad \text{(eq.15)}$$

For simulations in **Figures 1C** and **1F** and **Figures S5C** and S5D, we used the penalty values from Todorov (2005) for each state and control variable: w_p= 0. 00001, w_p = 2, w_v = 0. 2, w_f = 0. 02. The simulated speed and acceleration profiles are based on the optimal movement Time T_{opt} from **Figure S6B**. From **Figures S5C** and S5D, we obtain the peak speed, peak acceleration, and their respective times.

In summary, we apply the movement utility theory to calculate the optimal movement time for a simplified two-joint arm moving in a 2D plane. This movement time is primarily determined by the effective mass and temporal discounting of reward. Next, we employ the optimal control theory to model the kinematics of hand movement, including its position, speed, and acceleration. This approach also enables us to estimate the peak speed and acceleration, as well as their respective times, for reaches performed toward different target directions. To simulate the effect of microgravity, we introduce a mass underestimation factor to account for the kinematic changes associated with this perceptual error.

References

Berger M., Mescheriakov S., Molokanova E., Lechner-Steinleitner S., Seguer N., Kozlovskaya I. (1997) **Pointing arm movements in short- and long-term spaceflights** *Aviation, Space, and Environmental Medicine* **68**:781–787 Google Scholar

Berniker M., Kording K. (2008) **Estimating the sources of motor errors for adaptation and generalization** *Nature Neuroscience* **11**:1454–1461 Google Scholar

Bock O. (1998) **Problems of sensorimotor coordination in weightlessness** *Brain Research* **28**:155–160 Google Scholar

Bock O., Arnold K. E., Cheung B. S. (1996a) **Performance of a simple aiming task in hypergravity: II. detailed response characteristics** *Aviation, Space, and Environmental Medicine* **67**:133–138 Google Scholar

Bock O., Arnold K. E., Cheung B. S. (1996b) **Performance of a simple aiming task in hypergravity: I. overall accuracy** *Aviation, Space, and Environmental Medicine* **67**:127–132 Google Scholar

Bock O., Fowler B., Comfort D. (2001) **Human sensorimotor coordination during spaceflight: an analysis of pointing and tracking responses during the "Neurolab" Space Shuttle mission** *Aviation, Space, and Environmental Medicine* **72**:877–883 Google Scholar

Bock O., Fowler B., Jüngling S., Comfort D. (2003) **Visual-motor coordination during spaceflight** In: *The Neurolab Space Mission: Neuroscience Research in Space* pp. 83–89 Google Scholar

Bock O., Howard I. P., Money K. E., Arnold K. E. (1992) **Accuracy of aimed arm movements in changed gravity** *Aviation, Space, and Environmental Medicine* **63**:994–998 Google Scholar

Bock O., Weigelt C., Bloomberg J. J. (2010) **Cognitive demand of human sensorimotor performance during an extended space mission: a dual-task study** *Aviation, Space, and Environmental Medicine* **81**:819–824 Google Scholar

Brooks V. B. (1986) **The neural basis of motor control** Oxford: Oxford University Press Google Scholar

Chua R., Elliott D. (1993) **Visual regulation of manual aiming** *Human Movement Science* **12**:365–401 Google Scholar

Craik K. J. W. (1947) **Theory of the human operator in control systems; the operator as an engineering system** *The British Journal of Psychology* **38**:56–61 Google Scholar

Crevecoeur F., McIntyre J., Thonnard J.-L., Lefèvre P. (2010) **Movement stability under uncertain internal models of dynamics** *Journal of Neurophysiology* **104**:1301–1313 Google Scholar

Crevecoeur F., McIntyre J., Thonnard J.-L., Lefèvre P. (2014) **Gravity-dependent estimates of object mass underlie the generation of motor commands for horizontal limb movements**

Journal of Neurophysiology **112**:384–392 Google Scholar

Dimitriou M., Wolpert D. M., Franklin D. W. (2013) **The temporal evolution of feedback gains rapidly update to task demands** *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience* **33**:10898–10909 Google Scholar

Eddy D. R., Schiflett S. G., Schlegel R. E., Shehab R. L. (1998) **Cognitive performance aboard the life and microgravity spacelab** *Acta Astronautica* **43**:193–210 Google Scholar

Elliott D., Hansen S., Grierson L. E. M., Lyons J., Bennett S. J., Hayes S. J. (2010) **Goal-directed aiming: two components but multiple processes** *Psychological Bulletin* **136**:1023–1044 Google Scholar

Elliott D., Lyons J., Hayes S. J., Burkitt J. J., Roberts J. W., Grierson L. E. M., Hansen S., Bennett S. J. (2017) **The multiple process model of goal-directed reaching revisited** *Neuroscience and Biobehavioral Reviews* **72**:95–110 Google Scholar

Fercho K., Baugh L. A. (2014) **It's too quick to blame myself—the effects of fast and slow rates of change on credit assignment during object lifting** *Frontiers in Human Neuroscience* **8** https://doi.org/10.3389/fnhum.2014.00554 | Google Scholar

Flanagan J. R., Wing A. M. (1997) **The role of internal models in motion planning and control: evidence from grip force adjustments during movements of hand-held loads** *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience* **17**:1519–1528 Google Scholar

Flash T., Hogan N. (1985) **The coordination of arm movements: an experimentally confirmed mathematical model** *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience* **5**:1688–1703 Google Scholar

Fowler B., Meehan S., Singhal A. (2008) **Perceptual-motor performance and associated kinematics in space** *Human Factors* **50**:879–892 Google Scholar

Heuer H., Manzey D., Lorenz B., Sangals J. (2003) **Impairments of manual tracking performance during spaceflight are associated with specific effects of microgravity on visuomotor transformations** *Ergonomics* **46**:920–934 Google Scholar

Johansson R., Edin B. (1993) **Predictive feed-forward sensory control during grasping and manipulation in man** *Biomedical Research-Tokyo* **14**:95–106 Google Scholar

Kanas N., Manzey D. (2008) **Basic Issues of Human Adaptation to Space Flight** In: *Space Psychology and Psychiatry* Springer Netherlands pp. 15–48 Google Scholar

Kawato M. (1999) **Internal models for motor control and trajectory planning** *Current Opinion in Neurobiology* **9**:718–727 Google Scholar

Kirk R. (1968) **Experimental design: Procedures for the behavioral sciences** Brooks/Cole Google Scholar

Kluzik J., Diedrichsen J., Shadmehr R., Bastian A. J. (2008) **Reach adaptation: what determines whether we learn an internal model of the tool or adapt the model of our arm?** *Journal of Neurophysiology* **100**:1455–1464 Google Scholar

Kong G., Zhou Z., Wang Q., Kording K., Wei K. (2017) **Credit assignment between body and object probed by an object transportation task** *Scientific Reports* **7**:13415 Google Scholar

Krakauer J. W., Hadjiosif A. M., Xu J., Wong A. L., Haith A. M. (2019) **Motor learning** *Comprehensive Physiology* **9**:613–663 Google Scholar

Krakauer J. W., Pine Z. M., Ghilardi M. F., Ghez C. (2000) **Learning of visuomotor transformations for vectorial planning of reaching trajectories** *Journal of Neuroscience* **20**:8916–8924 Google Scholar

Kubis J. F., Mclaughlin E. J., Jackson J. M., Rusnak R., Mcbride G. H., Saxon S. V. (1977) **Task and work performance on Skylab missions 2, 3, and 4: Time and motion study: Experiment M151** *NASA. JSC Biomed. Results from Skylab* https://ntrs.nasa.gov/citations/19770026852

Kurtzer I. L., Pruszynski J. A., Scott S. H. (2008) **Long-latency reflexes of the human arm reflect an internal model of limb dynamics** *Current Biology: CB* **18**:449–453 Google Scholar

Lackner J. R., DiZio P. (1992) **Gravitoinertial force level affects the appreciation of limb position during muscle vibration** *Brain Research* **592**:175–180 Google Scholar

Lackner J. R., Dizio P. (1994) **Rapid adaptation to Coriolis force perturbations of arm trajectory** *Journal of Neurophysiology* **72**:299–313 Google Scholar

Layne C. S., Mulavara A. P., McDonald P. V., Pruett C. J., Kozlovskaya I. B., Bloomberg J. J. (2001) **Effect of long-duration spaceflight on postural control during self-generated perturbations** *Journal of Applied Physiology* **90**:997–1006 Google Scholar

Li W., Todorov E. (2004) **Iterative linear quadratic regulator design for nonlinear biological movement systems** In: *First International Conference on Informatics* https://www.scitepress.org/PublishedPapers/2004/11439/ | Google Scholar

Manzey D., Lorenz B., Poljakov V. (1998) **Mental performance in extreme environments: results from a performance monitoring study during a 438-day spaceflight** *Ergonomics* **41**:537–559 Google Scholar

Marsden C. D., Merton P. A., Morton H. B. (1976) **Stretch reflex and servo action in a variety of human muscles** *The Journal of Physiology* **259**:531–560 Google Scholar

McIntyre J., Zago M., Berthoz A., Lacquaniti F. (2001) **Does the brain model Newton's laws?** *Nature Neuroscience* **4**:693–694 Google Scholar

Mechtcheriakov S., Berger M., Molokanova E., Holzmueller G., Wirtenberger W., Lechner-Steinleitner S., De Col C., Kozlovskaya I., Gerstenbrand F. (2002) **Slowing of human arm movements during weightlessness: the role of vision** *European Journal of Applied Physiology* **87**:576–583 Google Scholar

Meyer D. E., Abrams R. A., Kornblum S., Wright C. E., Smith J. E. (1988) **Optimality in human motor performance: ideal control of rapid aimed movements** *Psychological Review* **95**:340–370 Google Scholar

Meyer D. E., Keith Smith J., Kornblum S., Abrams R., Wright C. (2018) **Speed—accuracy tradeoffs in aimed movements: Toward a theory of rapid voluntary action** In: *Attention and Performance XIII* https://doi.org/10.4324/9780203772010-6 | Google Scholar

Mussa-Ivaldi F. A., Hogan N., Bizzi E. (1985) **Neural, mechanical, and geometric factors subserving arm posture in humans** *The Journal of Neuroscience: The Official Journal of the Society for Neuroscience* **5**:2732–2743 Google Scholar

Newman D. J., Lathan C. E. (1999) **Memory processes and motor control in extreme environments** *IEEE Trans Syst Man Cybern C Appl Rev* **29**:387–394 Google Scholar

Paloski W. H., Reschke M. F., Black F. O., Dow R. S. (1999) **Recovery of postural equilibrium control following space flight (No. DSO-605)** ntrs.nasa.gov https://ntrs.nasa.gov/citations/20040201535 | Google Scholar

Papaxanthis C., Pozzo T., McIntyre J. (2005) **Kinematic and dynamic processes for the control of pointing movements in humans revealed by short-term exposure to microgravity** *Neuroscience* **135**:371–383 Google Scholar

Papaxanthis C., Pozzo T., Popov K. E., McIntyre J. (1998) **Hand trajectories of vertical arm movements in one-G and zero-G environments. Evidence for a central representation of gravitational force** *Experimental Brain Research* **120**:496–502 Google Scholar

Plamondon R., Alimi A. M. (1997) **Speed/accuracy trade-offs in target-directed movements** *The Behavioral and Brain Sciences* **20**:279–303 Google Scholar

Proske U., Gandevia S. C. (2012) **The proprioceptive senses: their roles in signaling body shape, body position and movement, and muscle force** *Physiological Reviews* **92**:1651–1697 Google Scholar

Proske U., Weber B. M. (2023) **Proprioceptive disturbances in weightlessness revisited** *NPJ Microgravity* **9**:64 Google Scholar

Reschke M. F., Bloomberg J. J., Harm D. L., Paloski W. H., Layne C., McDonald V. (1998) **Posture, locomotion, spatial orientation, and motion sickness as a function of space flight Brain Research** *Brain Research Reviews* **28**:102–117 Google Scholar

Rohrer B., Hogan N. (2006) **Avoiding spurious submovement decompositions II: a scattershot algorithm** *Biological Cybernetics* **94**:409–414 Google Scholar

Ross H. E., Reschke M. F. (1982) **Mass estimation and discrimination during brief periods of zero gravity** *Perception & Psychophysics* **31**:429–436 Google Scholar

Sangals J., Heuer H., Manzey D., Lorenz B. (1999) **Changed visuomotor transformations during and after prolonged microgravity** *Experimental Brain Research* **129**:378–390 Google Scholar

Shadmehr R., Huang H. J., Ahmed A. A. (2016) **A Representation of Effort in Decision-Making and Motor Control** *Current Biology: CB* **26**:1929–1934 Google Scholar

Shadmehr R., Mussa-Ivaldi F. A. (1994) **Adaptive representation of dynamics during learning of a motor task** *Journal of Neuroscience* **14**:3208–3224 Google Scholar

Smeets J. B., Brenner E. (1995) **Prediction of a moving target's position in fast goal-directed action** *Biological Cybernetics* **73**:519–528 Google Scholar

Strangman G. E., Sipes W., Beven G. (2014) **Human cognitive performance in spaceflight and analogue environments** *Aviation, Space, and Environmental Medicine* **85**:1033–1048 Google Scholar

Sutter K., Oostwoud Wijdenes L., van Beers R. J., Medendorp W. P. (2021) **Movement preparation time determines movement variability** *Journal of Neurophysiology* **125**:2375–2383 Google Scholar

Tafforin C., Thon B., Guell A., Campan R. (1989) **Astronaut behavior in an orbital flight situation: preliminary ethological observations** *Aviation, Space, and Environmental Medicine* **60**:949–956 Google Scholar

Tays G. D., Hupfeld K. E., McGregor H. R., Salazar A. P., De Dios Y. E., Beltran N. E., Reuter-Lorenz P. A., Kofman I. S., Wood S. J., Bloomberg J. J., Mulavara A. P., Seidler R. D. (2021) **The Effects of Long Duration Spaceflight on Sensorimotor Control and Cognition** *Frontiers in Neural Circuits* **15**:723504 Google Scholar

Tian Y., Zhang Z., Jiang C., Chen D., Liu Z., Wei M., Wang C., Wei K. (2024) **Stressors affect human motor timing during spaceflight** *NPJ Microgravity* in press Google Scholar

Todorov E. (2004) **Optimality principles in sensorimotor control** *Nature Neuroscience* **7**:907 Google Scholar

Todorov E. (2005) **Stochastic optimal control and estimation methods adapted to the noise characteristics of the sensorimotor system** *Neural Computation* **17**:1084–1108 Google Scholar

Wang J., Sainburg R. L. (2004) **Interlimb transfer of novel inertial dynamics is asymmetrical** *Journal of Neurophysiology* **92**:349–360 Google Scholar

Weber B., Riecke C., Stelzer M. (2019) **Disentangling the enigmatic slowing effect of microgravity on sensorimotor performance** In: *HFES-Europe* Google Scholar

Weber B., Stelzer M. (2022) **Sensorimotor impairments during spaceflight: Trigger mechanisms and haptic assistance** *Frontiers in Neuroergonomics* **3**:959894 Google Scholar

Wilke C., Synofzik M., Lindner A. (2013) **Sensorimotor recalibration depends on attribution of sensory prediction errors to internal causes** *PloS One* **8**:e54925 Google Scholar

Wolpert D. M., Diedrichsen J., Flanagan J. R. (2011) **Principles of sensorimotor learning** *Nature Reviews* **12**:739 Google Scholar

Wolpert D. M., Ghahramani Z., Jordan M. I. (1995) **An internal model for sensorimotor integration** *Science* **269**:1880–1882 Google Scholar

Wolpert D. M., Miall R. C. (1996) **Forward models for physiological motor control** *Neural Networks: The Official Journal of the International Neural Network Society* **9**:1265–1279 Google Scholar

Wolpert D. M., Miall R. C., Kawato M. (1998) **Internal models in the cerebellum** *Trends in Cognitive Sciences* **2**:338–347 Google Scholar

Woodworth R. S. (1899) **Accuracy of voluntary movement** *Psychological Monographs* **3**:i–114 Google Scholar

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Reviewer #1 (Public review):

Summary:

This article investigates the origin of movement slowdown in weightlessness by testing two possible hypotheses: the first is based on a strategic and conservative slowdown, presented as a scaling of the motion kinematics without altering its profile, while the second is based on the hypothesis of a misestimation of effective mass by the brain due to an alteration of gravity-dependent sensory inputs, which alters the kinematics following a controller parameterization error.

Strengths:

The article convincingly demonstrates that trajectories are affected in 0g conditions, as in previous work. It is interesting, and the results appear robust. However, I have two major reservations about the current version of the manuscript that prevent me from endorsing the conclusion in its current form.

Weaknesses:

(1) First, the hypothesis of a strategic and conservative slowdown implicitly assumes a similar cost function, which cannot be guaranteed, tested, or verified. For example, previous work has suggested that changing the ratio between the state and control weight matrices produced an alteration in movement kinematics similar to that presented here, without changing the estimated mass parameter (Crevecoeur et al., 2010, J Neurophysiol, 104 (3), 1301-1313). Thus, the hypothesis of conservative slowing cannot be rejected. Such a strategy could vary with effective mass (thus showing a statistical effect), but the possibility that the data reflect a combination of both mechanisms (strategic slowing and mass misestimation) remains open.

(2) The main strength of the article is the presence of directional effects expected under the hypothesis of mass estimation error. However, the article lacks a clear demonstration of such an effect: indeed, although there appears to be a significant effect of direction, I was not sure that this effect matched the model's predictions. A directional effect is not sufficient because the model makes clear quantitative predictions about how this effect should vary across directions. In the absence of a quantitative match between the model and the data, the authors' claims regarding the role of misestimating the effective mass remain unsupported.

In general, both the hypotheses of slowing motion (out of caution) and misestimating mass have been put forward in the past, and the added value of this article lies in demonstrating that the effect depended on direction. However, (1) a conservative strategy with a different cost function can also explain the data, and (2) the quantitative match between the directional effect and the model's predictions has not been established.

Specific points:

(1) I noted a lack of presentation of raw kinematic traces, which would be necessary to convince me that the directional effect was related to effective mass as stated.

(2) The presentation and justification of the model require substantial improvement; the reason for their presence in the supplementary material is unclear, as there is space to present the modelling work in detail in the main text. Regarding the model, some choices require justification: for example, why did the authors ignore the nonlinear Coriolis and centripetal terms?

(3) The increase in the proportion of trials with subcomponents is interesting, but the explanatory power of this observation is limited, as the initial percentage was already quite high (from 60-70% during the initial study to 70-85% in flight). This suggests that the potential effect of effective mass only explains a small increase in a trend already present in the initial study. A more critical assessment of this result is warranted.

https://doi.org/10.7554/eLife.107472.1.sa3

Reviewer #2 (Public review):

This study explores the underlying causes of the generalized movement slowness observed in astronauts in weightlessness compared to their performance on Earth. The authors argue that this movement slowness stems from an underestimation of mass rather than a deliberate reduction in speed for enhanced stability and safety.

Overall, this is a fascinating and well-written work. The kinematic analysis is thorough and comprehensive. The design of the study is solid, the collected dataset is rare, and the model tends to add confidence to the proposed conclusions. That being said, I have several comments that could be addressed to consolidate interpretations and improve clarity.

Main comments:

(1) Mass underestimation

a) While this interpretation is supported by data and analyses, it is not clear whether this gives a complete picture of the underlying phenomena. The two hypotheses (i.e., mass underestimation vs deliberate speed reduction) can only be distinguished in terms of velocity/acceleration patterns, which should display specific changes during the flight with a mass underestimation. The experimental data generally shows the expected changes but for the 45{degree sign} condition, no changes are observed during flight compared to the pre- and post-phases (Figure 4). In Figure 5E, only a change in the primary submovement peak velocity is observed for 45{degree sign}, but this finding relies on a more involved decomposition procedure. It suggests that there is something specific about 45{degree sign} (beyond its low effective mass). In such planar movements, 45{degree sign} often corresponds to a movement which is close to single-joint, whereas 90{degree sign} and 135{degree sign} involve multi-joint movements. If so, the increased proportion of submovements in 90{degree sign} and 135{degree sign} could indicate that participants had more difficulties in coordinating multi-joint movements during flight. Besides inertia, Coriolis and centripetal effects may be non-negligible in such fast planar reaching (Hollerbach & Flash, Biol Cyber, 1982) and, interestingly, they would also be affected by a mass underestimation (thus, this is not necessarily incompatible with the author's view; yet predicting the effects of a mass underestimation on Coriolis/centripetal torques would require a two-link arm model). Overall, I found the discrepancy between the 45{degree sign} direction and the other directions under-exploited in the current version of the article. In sum, could the corrective submovements be due to a misestimation of Coriolis/centripetal torques in the multi-joint dynamics (caused specifically -or not- by a mass underestimation)?

b) Additionally, since the taikonauts are tested after 2 or 3 weeks in flight, one could also assume that neuromuscular deconditioning explains (at least in part) the general decrease in movement speed. Can the authors explain how to rule out this alternative interpretation? For instance, weaker muscles could account for slower movements within a classical time-effort trade-off (as more neural effort would be needed to generate a similar amount of muscle force, thereby suggesting a purposive slowing down of movement). Therefore, could the observed results (slowing down + more submovements) be explained by some neuromuscular deconditioning combined with a difficulty in coordinating multi-joint movements in weightlessness (due to a misestimation or Coriolis/centripetal torques) provide an alternative explanation for the results?

(2) Modelling

a) The model description should be improved as it is currently a mix of discrete time and continuous time formulations. Moreover, an infinite-horizon cost function is used, but I thought the authors used a finite-horizon formulation with the prefixed duration provided by the movement utility maximization framework of Shadmehr et al. (Curr Biol, 2016). Furthermore, was the mass underestimation reflected both in the utility model and the optimal control model? If so, did the authors really compute the feedback control gain with the underestimated mass but simulate the system with the real mass? This is important because the mass appears both in the utility framework and in the LQ framework. Given the current interpretations, the feedforward command is assumed to be erroneous, and the feedback command would allow for motor corrections. Therefore, it could be clarified whether the feedback command also misestimates the mass or not, which may affect its efficiency. For instance, if both feedforward and feedback motor commands are based on wrong internal models (e.g., due to the mass underestimation), one may wonder how the astronauts would execute accurate goal-directed movements.

b) The model seems to be deterministic in its current form (no motor and sensory noise). Since the framework developed by Todorov (2005) is used, sensorimotor noise could have been readily considered. One could also assume that motor and sensory noise increase in microgravity, and the model could inform on how microgravity affects the number of submovements or endpoint variance due to sensorimotor noise changes, for instance.

c) Finally, how does the model distinguish the feedforward and feedback components of the motor command that are discussed in the paper, given that the model only yields a feedback control law? Does 'feedforward' refer to the motor plan here (i.e., the prefixed duration and arguably the precomputed feedback gain)?

(3) Brevity of movements and speed-accuracy trade-off

The tested movements are much faster (average duration approx. 350 ms) than similar self-paced movements that have been studied in other works (e.g., Wang et al., J Neurophysiology, 2016; Berret et al., PLOS Comp Biol, 2021, where movements can last about 900-1000 ms). This is consistent with the instructions to reach quickly and accurately, in line with a speed-accuracy trade-off. Was this instruction given to highlight the inertial effects related to the arm's anisotropy? One may however, wonder if the same results would hold for slower self-paced movements (are they also with reduced speed compared to Earth performance?). Moreover, a few other important questions might need to be addressed for completeness: how to ensure that astronauts did remember this instruction during the flight? (could the control group move faster because they better remembered the instruction?). Did the taikonauts perform the experiment on their own during the flight, or did one taikonaut assume the role of the experimenter?

(4) No learning effect

This is a surprising effect, as mentioned by the authors. Other studies conducted in microgravity have indeed revealed an optimal adaptation of motor patterns in a few dozen trials (e.g., Gaveau et al., eLife, 2016). Perhaps the difference is again related to single-joint versus multi-joint movements. This should be better discussed given the impact of this claim. Typically, why would a "sensory bias of bodily property" persist in microgravity and be a "fundamental constraint of the sensorimotor system"?

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Reviewer #3 (Public review):

Summary:

The authors describe an interesting study of arm movements carried out in weightlessness after a prolonged exposure to the so-called microgravity conditions of orbital spaceflight. Subjects performed radial point-to-point motions of the fingertip on a touch pad. The authors note a reduction in movement speed in weightlessness, which they hypothesize could be due to either an overall strategy of lowering movement speed to better accommodate the instability of the body in weightlessness or an underestimation of body mass. They conclude for the latter, mainly based on two effects. One, slowing in weightlessness is greater for movement directions with higher effective mass at the end effector of the arm. Two, they present evidence for an increased number of corrective submovements in weightlessness. They contend that this provides conclusive evidence to accept the hypothesis of an underestimation of body mass.

Strengths:

In my opinion, the study provides a valuable contribution, the theoretical aspects are well presented through simulations, the statistical analyses are meticulous, the applicable literature is comprehensively considered and cited, and the manuscript is well written.

Weaknesses:

Nevertheless, I am of the opinion that the interpretation of the observations leaves room for other possible explanations of the observed phenomenon, thus weakening the strength of the arguments.

First, I would like to point out an apparent (at least to me) divergence between the predictions and the observed data. Figures 1 and S1 show that the difference between predicted values for the 3 movement directions is almost linear, with predictions for 90º midway between predictions for 45º and 135º. The effective mass at 90º appears to be much closer to that of 45º than to that of 135º (Figure S1A). But the data shown in Figure 2 and Figure 3 indicate that movements at 90º and 135º are grouped together in terms of reaction time, movement duration, and peak acceleration, while both differ significantly from those values for movements at 45º.

Furthermore, in Figure 4, the change in peak acceleration time and relative time to peak acceleration between 1g and 0g appears to be greater for 90º than for 135º, which appears to me to be at least superficially in contradiction with the predictions from Figure S1. If the effective mass is the key parameter, wouldn't one expect as much difference between 90º and 135º as between 90º and 45º? It is true that peak speed (Figure 3B) and peak speed time (Figure 4B) appear to follow the ordering according to effective mass, but is there a mathematical explanation as to why the ordering is respected for velocity but not acceleration? These inconsistencies weaken the author's conclusions and should be addressed.

Then, to strengthen the conclusions, I feel that the following points would need to be addressed:

(1) The authors model the movement control through equations that derive the input control variable in terms of the force acting on the hand and treat the arm as a second-order low-pass filter (Equation 13). Underestimation of the mass in the computation of a feedforward command would lead to a lower-than-expected displacement to that command. But it is not clear if and how the authors account for a potential modification of the time constants of the 2nd order system. The CNS does not effectuate movements with pure torque generators. Muscles have elastic properties that depend on their tonic excitation level, reflex feedback, and other parameters. Indeed, Fisk et al.* showed variations of movement characteristics consistent with lower muscle tone, lower bandwidth, and lower damping ratio in 0g compared to 1g. Could the variations in the response to the initial feedforward command be explained by a misrepresentation of the limbs' damping and natural frequency, leading to greater uncertainty about the consequences of the initial command? This would still be an argument for unadapted feedforward control of the movement, leading to the need for more corrective movements. But it would not necessarily reflect an underestimation of body mass.

*Fisk, J. O. H. N., Lackner, J. R., & DiZio, P. A. U. L. (1993). Gravitoinertial force level influences arm movement control. Journal of neurophysiology, 69(2), 504-511.

(2) The movements were measured by having the subjects slide their finger on the surface of a touch screen. In weightlessness, the implications of this contact are expected to be quite different than those on the ground. In weightlessness, the taikonauts would need to actively press downward to maintain contact with the screen, while on Earth, gravity will do the work. The tangential forces that resist movement due to friction might therefore be different in 0g. This could be particularly relevant given that the effect of friction would interact with the limb in a direction-dependent fashion, given the anisotropy of the equivalent mass at the fingertip evoked by the authors. Is there some way to discount or control for these potential effects?

(3) The carefully crafted modelling of the limb neglects, nevertheless, the potential instability of the base of the arm. While the taikonauts were able to use their left arm to stabilize their bodies, it is not clear to what extent active stabilization with the contralateral limb can reproduce the stability of the human body seated in a chair in Earth gravity. Unintended motion of the shoulder could account for a smaller-than-expected displacement of the hand in response to the initial feedforward command and/or greater propensity for errors (with a greater need for corrective submovements) in 0g. The direction of movement with respect to the anchoring point could lead to the dependence of the observed effects on movement direction. Could this be tested in some way, e.g., by testing subjects on the ground while standing on an unstable base of support or sitting on a swing, with the same requirement to stabilize the torso using the contralateral arm?

The arguments for an underestimation of body mass would be strengthened if the authors could address these points in some way.

https://doi.org/10.7554/eLife.107472.1.sa1

Author response:

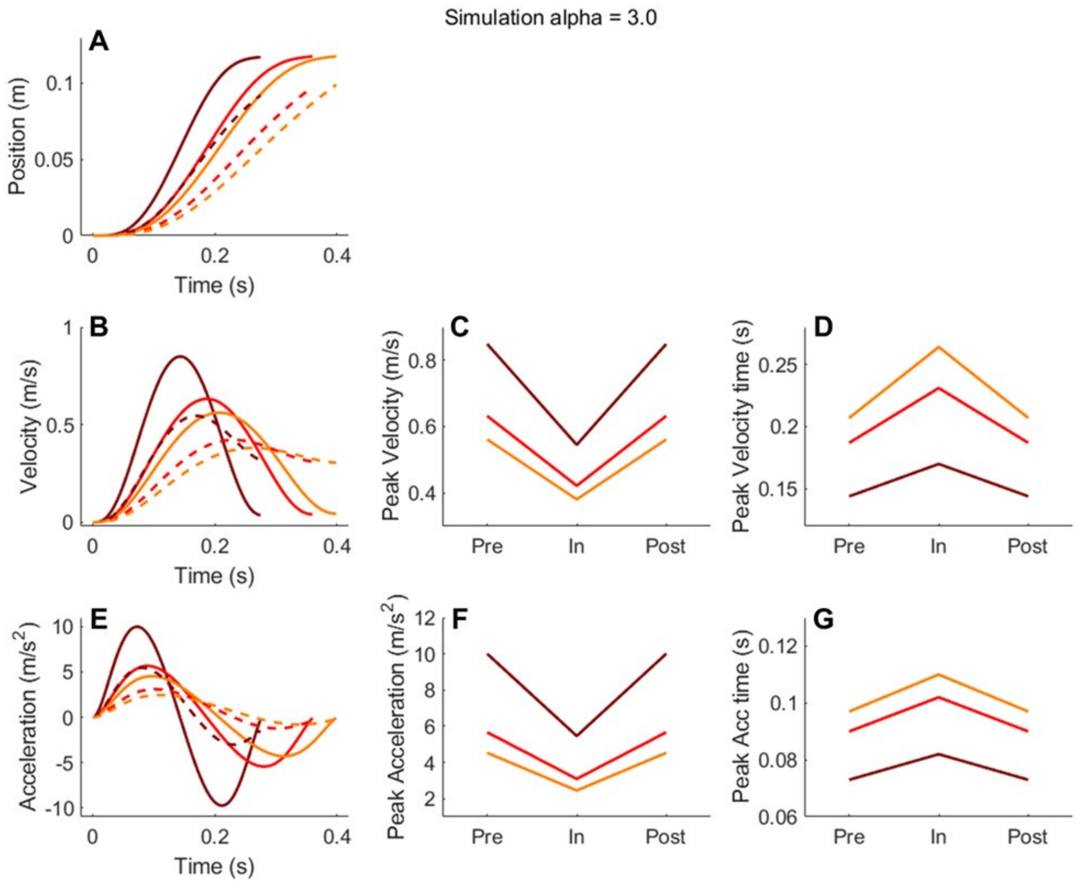
> ***Reviewer #1 (Public review):***
>
> *Summary:*

> *This article investigates the origin of movement slowdown in weightlessness by testing two possible hypotheses: the first is based on a strategic and conservative slowdown, presented as a scaling of the motion kinematics without altering its profile, while the second is based on the hypothesis of a misestimation of effective mass by the brain due to an alteration of gravity-dependent sensory inputs, which alters the kinematics following a controller parameterization error.*
>
> *Strengths:*
>
> *The article convincingly demonstrates that trajectories are affected in 0g conditions, as in previous work. It is interesting, and the results appear robust. However, I have two major reservations about the current version of the manuscript that prevent me from endorsing the conclusion in its current form.*
>
> *Weaknesses:*
>
> *(1) First, the hypothesis of a strategic and conservative slow down implicitly assumes a similar cost function, which cannot be guaranteed, tested, or verified. For example, previous work has suggested that changing the ratio between the state and control weight matrices produced an alteration in movement kinematics similar to that presented here, without changing the estimated mass parameter (Crevecoeur et al., 2010, J Neurophysiol, 104 (3), 1301-1313). Thus, the hypothesis of conservative slowing cannot be rejected. Such a strategy could vary with effective mass (thus showing a statistical effect), but the possibility that the data reflect a combination of both mechanisms (strategic slowing and mass misestimation) remains open.*

We test whether changing the ratio between the state and control weight matrices can generate the observed effect. As shown in Author response image 1 and Author response image 2, the cost function change cannot produce a reduced peak velocity/acceleration and their timing advance simultaneously, but a mass estimation change can. In other words, using mass underestimation alone can explain the two key findings, amplitude reduction and timing advance. Yes, we cannot exclude the possibility of a change in cost function on top of the mass underestimation, but the principle of Occam's Razor would support to adhering to a simple explanation, i.e., using body mass underestimation to explain the key findings. We will include our exploration on possible changes in cost function in the revision (in the Supplemental Materials).

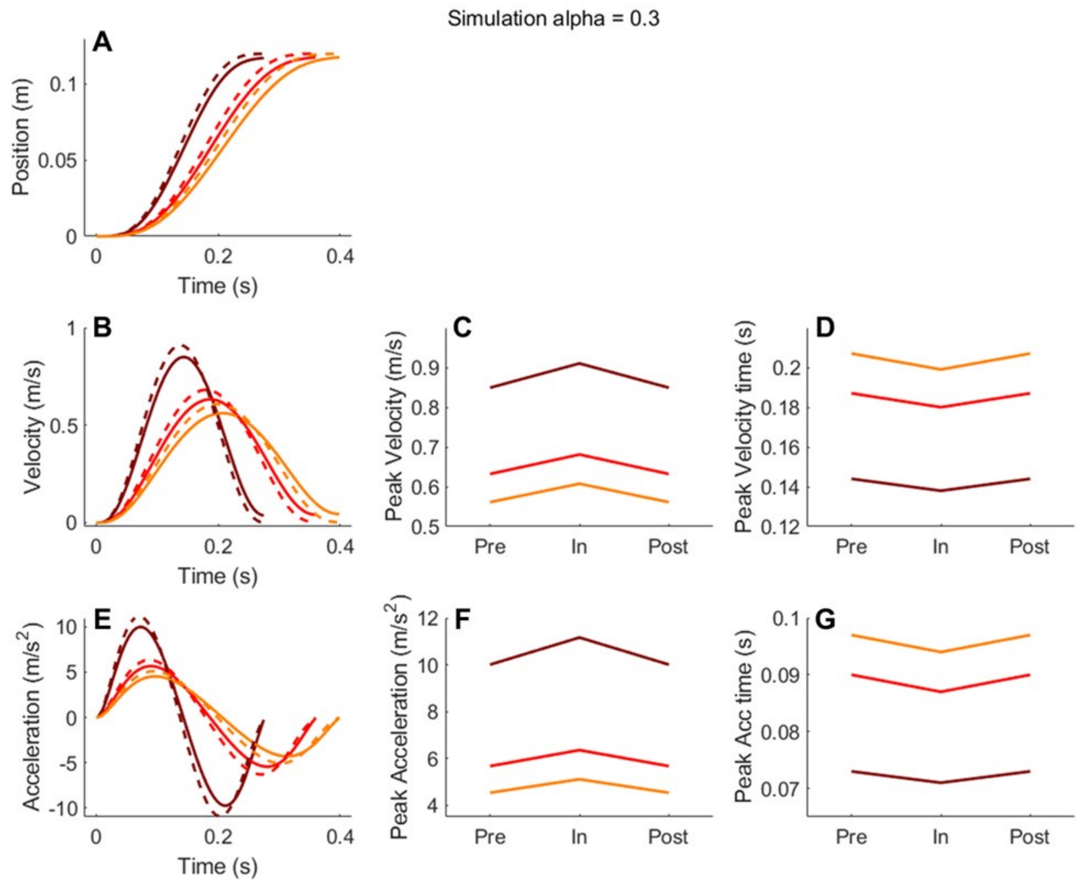
Author response image 1.

Simulation using an altered cost function with $\alpha = 3.0$. Panels A, B, and E show simulated position, velocity, and acceleration profiles, respectively, for the three movement directions. Solid lines correspond to pre- and post-exposure conditions, while dashed lines represent the in-flight condition. Panels C and D display the peak velocity and its timing across the three phases (Pre, In, Post), and Panels F and G show the corresponding peak acceleration and its timing. Note, varying the cost function, while leading to reduced peak velocity/acceleration, leads to an erroneous prediction of delayed timing of peak velocity/acceleration.



Author response image 2.

Simulation results using a cost function with α = 0.3. The format is the same as in Author response image 1. Note, this ten-fold decrease in α, while finally getting the timing of peak velocity/acceleration right (advanced or reduced), leads to an erroneous prediction of increased peak velocity/acceleration.



> *(2) The main strength of the article is the presence of directional effects expected under the hypothesis of mass estimation error. However, the article lacks a clear demonstration of such an effect: indeed, although there appears to be a significant effect of direction, I was not sure that this effect matched the model's predictions. A directional effect is not sufficient because the model makes clear quantitative predictions about how this effect should vary across directions. In the absence of a quantitative match between the model and the data, the authors' claims regarding the role of misestimating the effective mass remain unsupported.*

Our paper does not aim to quantitatively reproduce human reaching movements in microgravity. We will make this more clearly in the revision.

(1) The model is a simplification of the actual situation. For example, the model simulates an ideal case of moving a point mass (effective mass) without friction and without considering Coriolis and centripetal torques, while the actual situation is that people move their finger across a touch screen. The two-link arm model assumes planar movements, but our participants move their hand on a table top without vertical support to constrain their movement in 2D.

(2) Our study merely uses well-established (though simplified) models to qualitatively predict the overall behavioral patterns if mass underestimation is at play. For this purpose, the results are well in line with models' qualitative predictions: we indeed confirm that key kinematic features (peak velocity and acceleration) follow the same ranking order of movement direction conditions as predicted.

(3) Using model simulation to qualitatively predict human behavioral patterns is a common practice in motor control studies, prominent examples including the papers on optimal feedback control (Todorov, 2004 and 2005) and movement vigor (Shadmehr et al., 2016). In fact, our model was inspired by the model in the latter paper.

Citations:

Todorov, E. (2004). Optimality principles in sensorimotor control. Nature Neuroscience, 7(9), 907.

Todorov, E. (2005). Stochastic optimal control and estimation methods adapted to the noise characteristics of the sensorimotor system. Neural Computation, 17(5), 1084–1108.

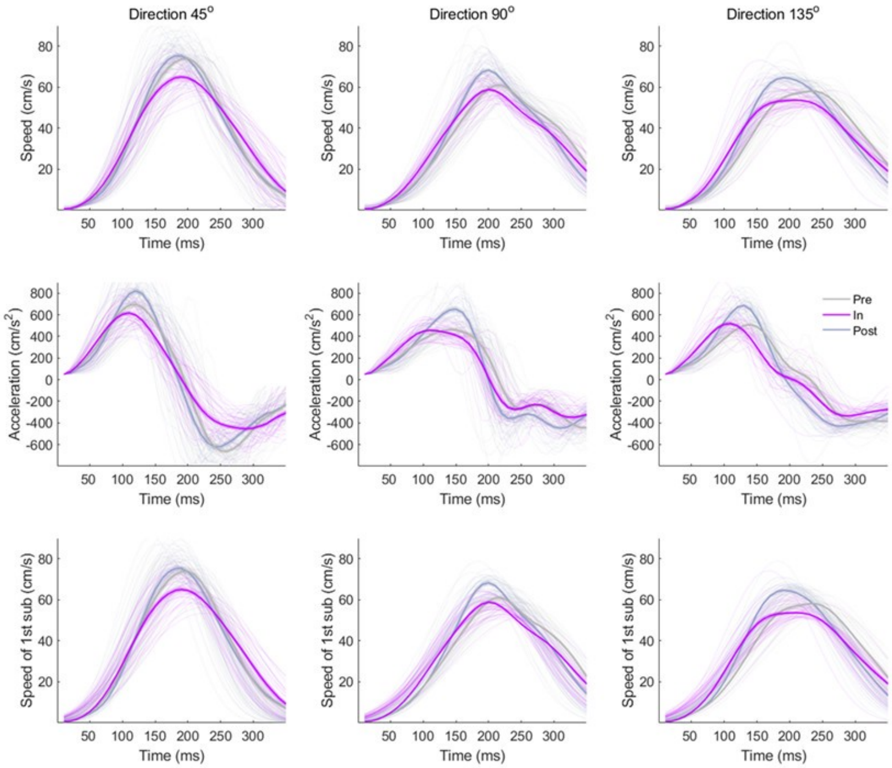
Shadmehr, R., Huang, H. J., & Ahmed, A. A. (2016). A Representation of Effort in Decision-Making and Motor Control. Current Biology: CB, 26(14), 1929–1934.

> *In general, both the hypotheses of slowing motion (out of caution) and misestimating mass have been put forward in the past, and the added value of this article lies in demonstrating that the effect depended on direction. However, (1) a conservative strategy with a different cost function can also explain the data, and (2) the quantitative match between the directional effect and the model's predictions has not been established.*
>
> *Specific points:*
>
> *(1) I noted a lack of presentation of raw kinematic traces, which would be necessary to convince me that the directional effect was related to effective mass as stated.*

We are happy to include exemplary speed and acceleration trajectories. One example subject's detailed trajectories are shown below and will be included in the revision. The reduced and advanced velocity/acceleration peaks are visible in typical trials.

Author response image 3.

Hand speed profiles (upper panels), hand acceleration profiles (middle panels) and speed profiles of the primary submovements (lower panels) towards different directions from an example participant.



> *(2) The presentation and justification of the model require substantial improvement; the reason for their presence in the supplementary material is unclear, as there is space to present the modelling work in detail in the main text. Regarding the model, some choices require justification: for example, why did the authors ignore the nonlinear Coriolis and centripetal terms?*

Response: In brief, our simulations show that Coriolis and centripetal forces, despite having some directional anisotropy, only have small effects on predicted kinematics (see our responses to Reviewer 2). We will move descriptions of the model into the main text with more justifications for using a simple model.

> *(3) The increase in the proportion of trials with subcomponents is interesting, but the explanatory power of this observation is limited, as the initial percentage was already quite high (from 60-70% during the initial study to 70-85% in flight). This suggests that the potential effect of effective mass only explains a small increase in a trend already present in the initial study. A more critical assessment of this result is warranted.*

Response: Indeed, the percentage of submovements only increases slightly, but the more important change is that the IPI (the inter-peak interval between submovements) also increases at the same time. Moreover, it is the effect of IPI that significantly predicts the duration increase in our linear mixed model. We will highlight this fact in our revision to avoid confusion.

Reviewer #2 (Public review):

This study explores the underlying causes of the generalized movement slowness observed in astronauts in weightlessness compared to their performance on Earth. The authors argue that this movement slowness stems from an underestimation of mass rather than a deliberate reduction in speed for enhanced stability and safety.

Overall, this is a fascinating and well-written work. The kinematic analysis is thorough and comprehensive. The design of the study is solid, the collected dataset is rare, and the model tends to add confidence to the proposed conclusions. That being said, I have several comments that could be addressed to consolidate interpretations and improve clarity.

Main comments:

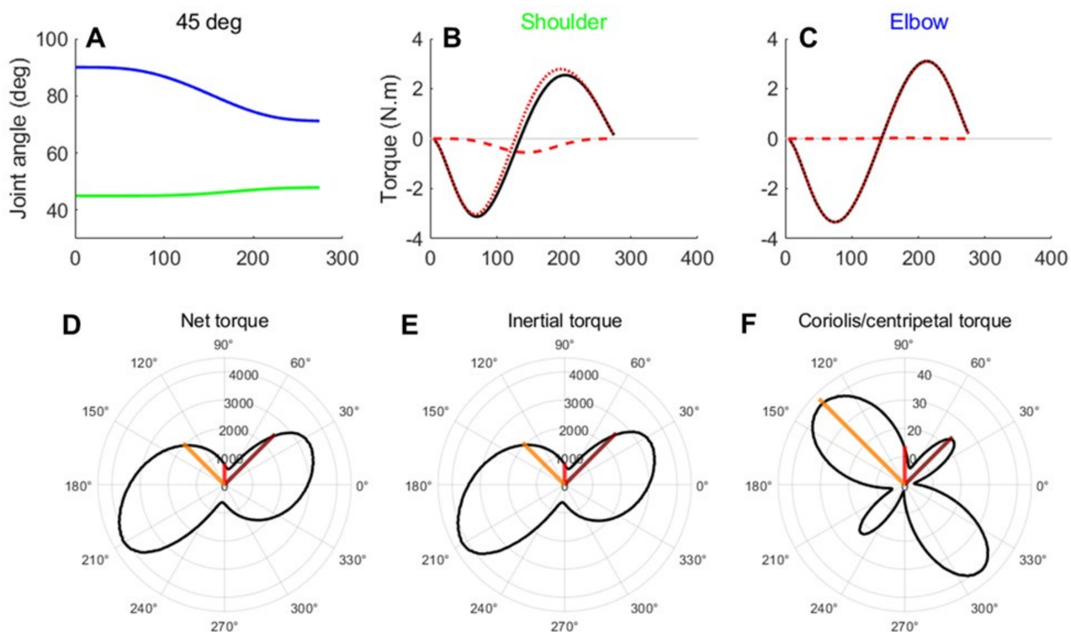
(1) Mass underestimation

a) While this interpretation is supported by data and analyses, it is not clear whether this gives a complete picture of the underlying phenomena. The two hypotheses (i.e., mass underestimation vs deliberate speed reduction) can only be distinguished in terms of velocity/acceleration patterns, which should display specific changes during the flight with a mass underestimation. The experimental data generally shows the expected changes but for the 45{degree sign} condition, no changes are observed during flight compared to the pre- and post-phases (Figure 4). In Figure 5E, only a change in the primary submovement peak velocity is observed for 45{degree sign}, but this finding relies on a more involved decomposition procedure. It suggests that there is something specific about 45{degree sign} (beyond its low effective mass). In such planar movements, 45{degree sign} often corresponds to a movement which is close to single-joint, whereas 90{degree sign} and 135{degree sign} involve multi-joint movements. If so, the increased proportion of submovements in 90{degree sign} and 135{degree sign} could indicate that participants had more difficulties in coordinating multi-joint movements during flight. Besides inertia, Coriolis and centripetal effects may be non-negligible in such fast planar reaching (Hollerbach & Flash, Biol Cyber, 1982) and, interestingly, they would also be affected by a mass underestimation (thus, this is not necessarily incompatible with the author's view; yet predicting the effects of a mass underestimation on Coriolis/centripetal torques would require a two-link arm model). Overall, I found the discrepancy between the 45{degree sign} direction and the other directions under-exploited in the current version of the article. In sum, could the corrective submovements be due to a misestimation of Coriolis/centripetal torques in the multi-joint dynamics (caused specifically -or not- by a mass underestimation)?

We agree that the effect of mass underestimation is less in the 45° direction than the other two directions, possibly related to its reliance on single-joint (elbow) as opposed to two-joints (elbow and shoulder) movements. Plus, movement correction using one joint is probably easier (as also suggested by another reviewer), this possibility will be further discussed in the revision. However, we find that our model simplification (excluding Coriolis and centripetal torques) does not affect our main conclusions at all. First, we performed a simple simulation and found that, under the current optimal hand trajectory, incorporating Coriolis and centripetal torques has only a limited impact on the resulting joint torques (see simulations in Author response image 4). One reason is that we used smaller movements than Hallerbach & Flash did. In addition, we applied an optimal feedback control model to a more realistic 2-joint arm configuration. Despite its simplicity, this model produced a speed profile consistent with our current predictions and made similar predictions regarding the effects of mass underestimation (Author response image 5). We will provide a more realistic 2-joint arm model muscle dynamics in the revision to improve the simulation further, but the message will be same: including or excluding Coriolis and centripetal torques will not affect the theoretical predictions about mass underestimation. Second, as the reviewer correctly pointed out, the mass (and its underestimation) also affects these two torque terms, thus its effect on kinematic measures is not affected much even with the full model.

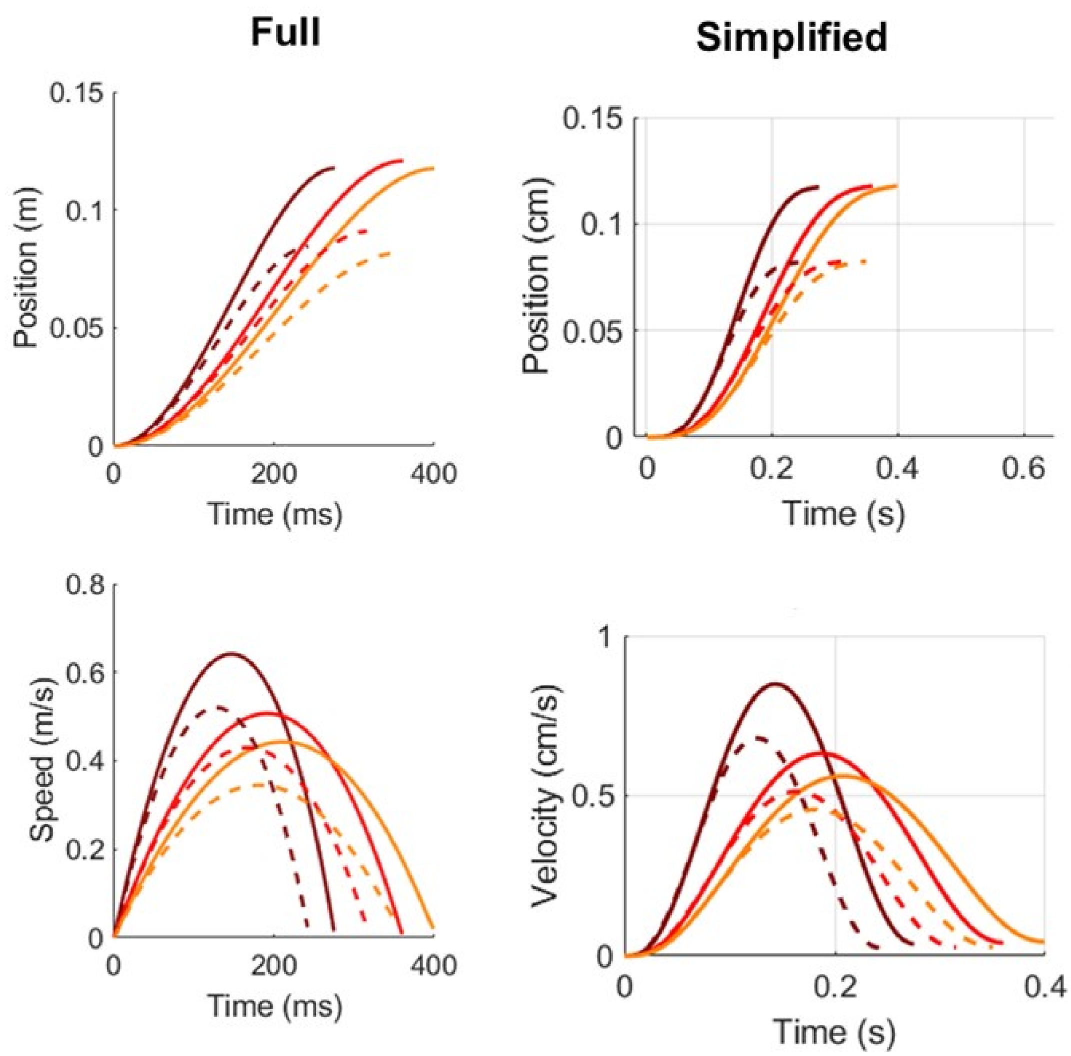
Author response image 4.

Joint angles and joint torque of shoulder and elbow with simulated trajectories towards different directions. A. Shoulder (green) and elbow (blue) angles change with time for the 45° movement direction. B. Components of joint interaction torques at the shoulder. Solid line: net torque at the shoulder; dotted line: shoulder inertia torque; dashed line: shoulder Coriolis and centripetal torque. C. Same plot as B for the elbow joint. D–F. Coriolis and centripetal components in the full 360° workspace, beyond three movement directions (45°, 90°, and 135°). D. Net torque. E. Inertial torque. F. Combined Coriolis and centripetal torque. Note the polar plots of Coriolis/centripetal torques (F) have a scale that is two magnitudes smaller than that of inertial torque in our simulation. All torques were simulated with the optimal movement duration. Torques were squared and integrated over each trajectory.



Author response image 5.

Comparison between simulation results from the full model with the addition of Coriolis/centripetal torques (left) and the simplified model (right). The position profiles (top) and the corresponding speed profiles low) are shown. Solid lines are for normal mass estimation and dashed lines for mass underestimation in microgravity. The three colors represent three movement directions (dark red: 45°, red: 90°, yellow: 135°). The full model used a 2-link arm model without realistic muscle dynamics yet (will include in the formal revision) thus the speed profile is not smooth. Importantly, the full model also predict the same effect of mass underestimation, i.e., reduced peak velocity/acceleration and their timing advance.



> *b) Additionally, since the taikonauts are tested after 2 or 3 weeks in flight, one could also assume that neuromuscular deconditioning explains (at least in part) the general decrease in movement speed. Can the authors explain how to rule out this alternative interpretation? For instance, weaker muscles could account for slower movements within a classical time-effort trade-off (as more neural effort would be needed to generate a similar amount of muscle force, thereby suggesting a purposive slowing down of movement). Therefore, could the observed results (slowing down + more submovements) be explained by some neuromuscular deconditioning combined with a difficulty in coordinating multi-joint movements in weightlessness (due to a misestimation or Coriolis/centripetal torques) provide an alternative explanation for the results?*

Response: Neuromuscular deconditioning is indeed a space or microgravity effect; thanks for bringing this up as we omitted the discussion of its possible contribution in the initial submission. However, muscle weakness is less for upper-limb muscles than for postural and lower-limb muscles (Tesch et al., 2005). The handgrip strength decreases 5% to 15% after several months (Moosavi et al., 2021); shoulder and elbow muscles atrophy, though not directly measured, was estimated to be minimal (Shen et al., 2017). The muscle weakness is unlikely to play a major role here since our reaching task involves small movements (~12cm) with joint torques of a magnitude of ~2N·m. Coriolis/centripetal torques does not affect the putative mass effect (as shown above simulations). The reviewer suggests that their poor coordination in microgravity might contribute to slowing down + more submovements. Poor coordination is an umbrella term for any motor control problems, and it can explain any microgravity effect. The feedforward control changes caused by mass underestimation can also be viewed as poor coordination. If we limit it as the coordination of the two joints or coordinating Coriolis/centripetal torques, we should expect to see some trajectory curvature changes in microgravity. However, we further analyzed our reaching trajectories and found no sign of curvature increase in our large collection of reaching movements. We probably have the largest dataset of reaching movements collected in microgravity thus far, given that we had 12 taikonauts and each of them performed about 480 to 840 reaching trials during their spaceflight. We believe the probability of Type II error is quite low here. We will include descriptive statistics of these new analyses in our revision.

Citation: Tesch, P. A., Berg, H. E., Bring, D., Evans, H. J., & LeBlanc, A. D. (2005). Effects of 17-day spaceflight on knee extensor muscle function and size. European journal of applied physiology, 93(4), 463-468.

Moosavi, D., Wolovsky, D., Depompeis, A., Uher, D., Lennington, D., Bodden, R., & Garber, C. E. (2021). The effects of spaceflight microgravity on the musculoskeletal system of humans and animals, with an emphasis on exercise as a countermeasure: A systematic scoping review. Physiological Research, 70(2), 119.

Shen, H., Lim, C., Schwartz, A. G., Andreev-Andrievskiy, A., Deymier, A. C., & Thomopoulos, S. (2017). Effects of spaceflight on the muscles of the murine shoulder. The FASEB Journal, 31(12), 5466.

> *(2) Modelling*
>
> *a) The model description should be improved as it is currently a mix of discrete time and continuous time formulations. Moreover, an infinite-horizon cost function is used, but I thought the authors used a finite-horizon formulation with the prefixed duration provided by the movement utility maximization framework of Shadmehr et al. (Curr Biol, 2016). Furthermore, was the mass underestimation reflected both in the utility model and the optimal control model? If so, did the authors really compute the feedback control gain with the underestimated mass but simulate the system with the real mass? This is important because the mass appears both in the utility framework and in the LQ framework. Given the current interpretations, the feedforward command is assumed to be erroneous, and the feedback command would allow for motor corrections. Therefore, it could be clarified whether the feedback command also misestimates the mass or not, which may affect its efficiency. For instance, if both feedforward and feedback motor commands are based on wrong internal models (e.g., due to the mass underestimation), one may wonder how the astronauts would execute accurate goal-directed movements.*
>
> *b) The model seems to be deterministic in its current form (no motor and sensory noise). Since the framework developed by Todorov (2005) is used, sensorimotor noise could have been readily considered. One could also assume that motor and sensory noise increase in microgravity, and the model could inform on how microgravity affects the number of submovements or endpoint variance due to sensorimotor noise changes, for instance.*
>
> *c) Finally, how does the model distinguish the feedforward and feedback components of the motor command that are discussed in the paper, given that the model only yields a feedback control law? Does 'feedforward' refer to the motor plan here (i.e., the prefixed duration and arguably the precomputed feedback gain)?*

We appreciate these very helpful suggestions about our model presentation. Indeed, our initial submission did not give detailed model descriptions in the main text, due to text limits for early submissions. We actually used a finite-horizon framework throughout, with a pre-specified duration derived from the utility model. In the revision, we will make that point clear, and we will also revise the Methods section to explicitly distinguish feedforward vs. feedback components, clarify the use of mass underestimation in both utility and control models, and update the equations accordingly.

> *(3) Brevity of movements and speed-accuracy trade-off*
>
> *The tested movements are much faster (average duration approx. 350 ms) than similar self-paced movements that have been studied in other works (e.g., Wang et al., J Neurophysiology, 2016; Berret et al., PLOS Comp Biol, 2021, where movements can last about 900-1000 ms). This is consistent with the instructions to reach quickly and accurately, in line with a speed-accuracy trade-off. Was this instruction given to highlight the inertial effects related to the arm's anisotropy? One may however, wonder if the same results would hold for slower self-paced movements (are they also with reduced speed compared to Earth performance?). Moreover, a few other important questions might need to be addressed for completeness: how to ensure that astronauts did remember this instruction during the flight? (could the control group move faster because they better remembered the instruction?). Did the taikonauts perform the experiment on their own during the flight, or did one taikonaut assume the role of the experimenter?*

Thanks for highlighting the brevity of movements in our experiment. Our intention in emphasizing fast movements is to rigorously test whether movement is indeed slowed down in microgravity. The observed prolonged movement duration clearly shows that microgravity affects people's movement duration, even when they are pushed to move fast. The second reason for using fast movement is to highlight that feedforward control is affected in microgravity. Mass underestimation specifically affects feedforward control in the first place. Slow movement would inevitably have online corrections that might obscure the effect of mass underestimation. Note that movement slowing is not only observed in our speed-emphasized reaching task, but also in whole-arm pointing in other astronauts studies (Berger, 1997; Sangals, 1999), which have been quoted in our paper. We thus believe these findings are generalizable.

Regarding the consistency of instructions: all our experiments conducted in the Tiangong space station were monitored in real time by experimenters in the Control Center located in Beijing. The task instructions were presented on the initial display of the data acquisition application and ample reading time was allowed. In fact, all the pre-, in-, and post-flight test sessions were administered by the same group of experimenters with the same instruction. It is common that astronauts serve both as participants and experimenters at the same time. And, they were well trained for this type of role on the ground. Note that we had multiple pre-flight test sessions to familiarize them with the task. All these rigorous measures were in place to obtain high-quality data. We will include these experimental details and the rationales for emphasizing fast movements in the revision.

Citations:

Berger, M., Mescheriakov, S., Molokanova, E., Lechner-Steinleitner, S., Seguer, N., & Kozlovskaya, I. (1997). Pointing arm movements in short- and long-term spaceflights. Aviation, Space, and Environmental Medicine, 68(9), 781–787.

Sangals, J., Heuer, H., Manzey, D., & Lorenz, B. (1999). Changed visuomotor transformations during and after prolonged microgravity. Experimental Brain Research. Experimentelle Hirnforschung. Experimentation Cerebrale, 129(3), 378–390.

> *(4) No learning effect*

This is a surprising effect, as mentioned by the authors. Other studies conducted in microgravity have indeed revealed an optimal adaptation of motor patterns in a few dozen trials (e.g., Gaveau et al., eLife, 2016). Perhaps the difference is again related to single-joint versus multi-joint movements. This should be better discussed given the impact of this claim. Typically, why would a "sensory bias of bodily property" persist in microgravity and be a "fundamental constraint of the sensorimotor system"?

We believe the differences between our study and Gaveau et al.'s study cannot be simply attributed to single-joint versus multi-joint movements. One of the most salient differences is that their adaptation is about incorporating microgravity in control for minimizing effort, while our adaptation is about rightfully perceiving body mass. We will elaborate on possible reasons for the lack of learning in the light of this previous study.

We can elaborate on "sensory bias" and "fundamental constraint of the sensorimotor system". If an inertial change is perceived (like an extra weight attached to the forearm, as in previous motor adaptation studies), people can adapt their reaching in tens of trials. In this case, sensory cues are veridical as they correctly inform about the inertial perturbation. However, in microgravity, reduced gravitational pull and proprioceptive inputs constantly inform the controller that the body mass is less than its actual magnitude. In other words, sensory cues in space are misleading for estimating body mass. The resulting sensory bias prevents the sensorimotor system from correctly adapt. Our statement was too brief in the initial submission; we will expand it in the revision.

Reviewer #3 (Public review):

Summary:

The authors describe an interesting study of arm movements carried out in weightlessness after a prolonged exposure to the so-called microgravity conditions of orbital spaceflight. Subjects performed radial point-to-point motions of the fingertip on a touch pad. The authors note a reduction in movement speed in weightlessness, which they hypothesize could be due to either an overall strategy of lowering movement speed to better accommodate the instability of the body in weightlessness or an underestimation of body mass. They conclude for the latter, mainly based on two effects. One, slowing in weightlessness is greater for movement directions with higher effective mass at the end effector of the arm. Two, they present evidence for an increased number of corrective submovements in weightlessness. They contend that this provides conclusive evidence to accept the hypothesis of an underestimation of body mass.

Strengths:

In my opinion, the study provides a valuable contribution, the theoretical aspects are well presented through simulations, the statistical analyses are meticulous, the applicable literature is comprehensively considered and cited, and the manuscript is well written.

Weaknesses:

Nevertheless, I am of the opinion that the interpretation of the observations leaves room for other possible explanations of the observed phenomenon, thus weakening the strength of the arguments.

First, I would like to point out an apparent (at least to me) divergence between the predictions and the observed data. Figures 1 and S1 show that the difference between predicted values for the 3 movement directions is almost linear, with predictions for 90º midway between predictions for 45º and 135º. The effective mass at 90º appears to be

> *much closer to that of 45º than to that of 135º (Figure S1A). But the data shown in Figure 2 and Figure 3 indicate that movements at 90º and 135º are grouped together in terms of reaction time, movement duration, and peak acceleration, while both differ significantly from those values for movements at 45º.*
>
> *Furthermore, in Figure 4, the change in peak acceleration time and relative time to peak acceleration between 1g and 0g appears to be greater for 90º than for 135º, which appears to me to be at least superficially in contradiction with the predictions from Figure S1. If the effective mass is the key parameter, wouldn't one expect as much difference between 90º and 135º as between 90º and 45º? It is true that peak speed (Figure 3B) and peak speed time (Figure 4B) appear to follow the ordering according to effective mass, but is there a mathematical explanation as to why the ordering is respected for velocity but not acceleration? These inconsistencies weaken the author's conclusions and should be addressed.*

Indeed, the model predicts an almost equal separation between 45° and 90° and between 90° and 135°, while the data indicate that the spacing between 45° and 90° is much smaller than between 90° and 135°. We do not regard the divergence as evidence undermining our main conclusion since 1) the model is a simplification of the actual situation. For example, the model simulates an ideal case of moving a point mass (effective mass) without friction and without considering Coriolis and centripetal torques. 2) Our study does not make quantitative predictions of all the key kinematic measures; that will require model fitting and parameter estimation; instead, our study uses well-established (though simplified) models to qualitatively predict the overall behavioral pattern we would observe. For this purpose, our results are well in line with our expectations: though we did not find equal spacing between direction conditions, we do confirm that the key kinematic properties (Figure 2 and Figure 3 as questioned) follow the same ranking order of directions as predicted.

We thank the reviewer for pointing out the apparent discrepancy between model simulation and observed data. We will elaborate on the reasons behind the discrepancy in the revision.

> *Then, to strengthen the conclusions, I feel that the following points would need to be addressed:*
>
> *(1) The authors model the movement control through equations that derive the input control variable in terms of the force acting on the hand and treat the arm as a second-order low-pass filter (Equation 13). Underestimation of the mass in the computation of a feedforward command would lead to a lower-than-expected displacement to that command. But it is not clear if and how the authors account for a potential modification of the time constants of the 2nd order system. The CNS does not effectuate movements with pure torque generators. Muscles have elastic properties that depend on their tonic excitation level, reflex feedback, and other parameters. Indeed, Fisk et al.* showed variations of movement characteristics consistent with lower muscle tone, lower bandwidth, and lower damping ratio in 0g compared to 1g. Could the variations in the response to the initial feedforward command be explained by a misrepresentation of the limbs' damping and natural frequency, leading to greater uncertainty about the consequences of the initial command? This would still be an argument for unadapted feedforward control of the movement, leading to the need for more corrective movements. But it would not necessarily reflect an underestimation of body mass.*
>
> **Fisk, J. O. H. N., Lackner, J. R., & DiZio, P. A. U. L. (1993). Gravitoinertial force level influences arm movement control. Journal of neurophysiology, 69(2), 504-511.*

We agree that muscle properties, tonic excitation level, proprioception-mediated reflexes all contribute to reaching control. Fisk et al. (1993) study indeed showed that arm movement kinematics change, possibly owing to lower muscle tone and/or damping. However, reduced muscle damping and reduced spindle activity are more likely to affect feedback-based movements. Like in Fisk et al.'s study, people performed *continuous* arm movements with eyes closed; thus their movements largely relied on proprioceptive control. Our major findings are about the feedforward control, i.e., the reduced and "advanced" peak velocity/acceleration in discrete and ballistic reaching movements. Note that the peak acceleration happens as early as approximately 90-100ms into the movements, clearly showing that feedforward control is affected -- a different effect from Fisk et al's findings. It is unlikely that people "advanced" their peak velocity/acceleration because they feel the need for more later corrective movements. Thus, underestimation of body mass remains the most plausible explanation.

> *(2) The movements were measured by having the subjects slide their finger on the surface of a touch screen. In weightlessness, the implications of this contact are expected to be quite different than those on the ground. In weightlessness, the taikonauts would need to actively press downward to maintain contact with the screen, while on Earth, gravity will do the work. The tangential forces that resist movement due to friction might therefore be different in 0g. This could be particularly relevant given that the effect of friction would interact with the limb in a direction-dependent fashion, given the anisotropy of the equivalent mass at the fingertip evoked by the authors. Is there some way to discount or control for these potential effects?*

We agree that friction might play a role here, but normal interaction with a touch screen typically involves friction between 0.1 and 0.5N (e.g., Ayyildiz et al., 2018). We believe that the directional variation is even smaller than 0.1N. It is very small compared to the force used to accelerate the arm for the reaching movement (10-15N). Thus, friction anisotropy is unlikely to explain our data.

Citation: Ayyildiz M, Scaraggi M, Sirin O, Basdogan C, Persson BNJ. Contact mechanics between the human finger and a touchscreen under electroadhesion. Proc Natl Acad Sci U S A. 2018 Dec 11;115(50):12668-12673.

> *(3) The carefully crafted modelling of the limb neglects, nevertheless, the potential instability of the base of the arm. While the taikonauts were able to use their left arm to stabilize their bodies, it is not clear to what extent active stabilization with the contralateral limb can reproduce the stability of the human body seated in a chair in Earth gravity. Unintended motion of the shoulder could account for a smaller-than-expected displacement of the hand in response to the initial feedforward command and/or greater propensity for errors (with a greater need for corrective submovements) in 0g. The direction of movement with respect to the anchoring point could lead to the dependence of the observed effects on movement direction. Could this be tested in some way, e.g., by testing subjects on the ground while standing on an unstable base of support or sitting on a swing, with the same requirement to stabilize the torso using the contralateral arm?*

Body stabilization is always a challenge for human movement studies in space. We minimized its potential confounding effects by using left-hand grasping and foot straps for postural support throughout the experiment. We would argue shoulder stability is an unlikely explanation because unexpected shoulder instability should not affect the feedforward (early) part of the ballistic reaching movement: the reduced peak acceleration and its early peak were observed at about 90-100ms after movement initiation. This effect is too early to be explained by an expected stability issue.

The arguments for an underestimation of body mass would be strengthened if the authors could address these points in some way.

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